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TOOTHED CVT FOR WHICH TEETH RE-ENGAGEMENT IS IMPROVED

Abstract

A toothed CVT using a Cone with One Torque Transmitting Member that is coupled by a Transmission Belt to another Cone with One Torque Transmitting Member for which teeth re-engagement between the Torque Transmitting Members and their Transmission Belt is improved. Said CVT uses a slack-side Tensioning Pulley in order to maintain the tension in the slack-side of the Transmission Belt almost constant so as to ensure smooth re-engagement of the Driven Cone; and a Transmission Diameter Compensating Mechanism, which increases the Transmission Diameter of the Driving Cone when the torque being pulled by the Driving Cone is increased, in order to compensate for an increase in “Transmission Belt Teeth Compression” and an increase in “stretching of the tense-side of the Transmission Belt”, due to an increase in tension in the tense-side of the Transmission Belt when an increased torque is transmitted; to ensure smooth re-engagement of the Driving Cone.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This invention is a Continuation of U.S. patent application Ser. No. 17/440,717 which was filed on 17 Sep. 2021.

BACKGROUND

Field of Invention

[0002] This invention relates to torque/speed transmissions, specifically to a Continuously Variable Transmission (CVT) that uses a “Cone with One Torque Transmitting Member” that is coupled by a Transmission Belt to another “Cone with One Torque Transmitting Member” for which the re-engagement of the teeth of the Torque Transmitting Members with teeth of their Transmission Belt is improved.

[0003] Said CVT uses a slack-side Tensioning Pulley in order to maintain the tension in the slack-side of the Transmission Belt almost constant so as to ensure smooth re-engagement of the Driven Cone; and a Transmission Diameter Compensating Mechanism, which increases the Transmission Diameter of the Driving Cone when the torque being pulled by the Driving Cone is increased, in order to compensate for an increase in “Transmission Belt Teeth Compression” and an increase in “stretching of the tense side of the Transmission Belt”, due to an increase in tension in the tense side of the Transmission Belt when an increased amount of torque is being transmitted by said CVT; to ensure smooth re-engagement of the Driving Cone.

[0004] Preferably said CVT also uses a “slack side Constrainer” that removes and/or provides “Transmission Belt slack” as needed when the axial position of the Cones are “aligned and “not aligned” for their current Transmission Ratio (see BRIEF SUMMARY OF THE INVENTION SECTION) without substantially changing the “angles at which the Transmission Belt approaches and leaves its Tensioning Pulley”; so as to reduce/minimize changes of the “angles at which the Transmission Belt approaches and leaves its Tensioning Pulley” due to whether the Cones are “aligned and “not aligned” for their current Transmission Ratio, and/or due to other operational variations of the CVT. This will reduce variation in the tension in the slack side of the Transmission Belt so as to further improve re-engagement of the Driven Cone. In addition, said “slack side Constrainer” also prevents teeth ratcheting at the point where engagement between the pulling Torque Transmitting Member and its Transmission Belt ends.

[0005] And preferably said CVT also uses a “Method of changing the Transmission Diameters of the Cones in increments specifically selected for each Transmission Diameter and Transmission Ratio” in order to compensate for: a) variation in “Transmission Belt Teeth Compression” at different Transmission Diameters; and b) variation in “slack side transmission belt tension” due to variation in the angle at which the Transmission Belt approaches and leaves its Tensioning Pulley based on whether the Cones are “aligned” or “not aligned” for their current Transmission Ratio.

Description of Prior Art

[0006] The inventions of this disclosure are primarily for a “cone with one torque transmitting member” described in U.S. Pat. No. 8,628,439.

[0007] A “cone with one torque transmitting member” of U.S. Pat. No. 8,628,439 (which is referred to as a front pin belt cone assembly 520A) is described in column 90 line 64 to column 99 line 11; and shown in FIGS. 91A, 91B, 92A, 92B, 93 to 101 of the Published Patent.

[0008] A “cone with one torque transmitting member” is basically a cone that has a “torque

transmitting member” that is rotatably fixed relative to the surface of its cone, but can slide axially relative to the surface of its cone. The “torque transmitting member” is used for torque transmission and can be made toothed. Preferably (although not absolutely necessary), a “cone with one torque transmitting member” also has a “non-torque transmitting member”, which is primarily used to maintain the alignment of the transmission belt of its cone and balance-out the centrifugal forces of the “torque transmitting member”. For the cone with one torque transmitting member of U.S. Pat. No. 8,628,439, which is referenced in the previous paragraph, the a “torque transmitting member” is labeled as “pin belt torque transmitting member **590**” and the “non-torque transmitting member” is labeled as “pin belt non-torque transmitting member **690**”.

[0009] An example of CVT that is constructed of out of two “cone with one torque transmitting member”, which are labeled as Cone **1A** and Cone **1B**, is shown in FIGS. **1** to **4**. This CVT comprises of Cone **1A** mounted on one spline that is coupled by a Transmission Belt **2** (which can be replaced with a chain in an alternate configuration) to a Cone **1B** mounted on another spline. A Tensioning Pulley **3**, positioned on the slack side of Transmission Belt **2**, is used to maintain proper tension in Transmission Belt **2** as the axial position of the cones are changed independent of each other. And a Support Pulley **4** is used to ensure that for each cone at least a portion of its torque transmitting member is engaged with Transmission Belt **2** for torque transmission at all times.

[0010] Cones **1A** and **1B** are each a “cone with one torque transmitting member”. Cone **1A** has a Torque Transmitting Member **1A-M1**, Non-Torque Transmitting Member **1A-M2**, and a Leveling Loop **1A-M3**. Cone **1B** has a Torque Transmitting Member **1B-M1**, Non-torque Transmitting Member **1B-M2**, and a Leveling Loop **1B-M3**. Torque transmitting members **1A-M1** and **1B-M1** have teeth so that toothed torque transmission can be used, although friction torque transmitting members can also be used.

[0011] A leveling loop, such as Leveling Loop **1A-M3** and Leveling Loop **1B-M3**, is a flexible loop with a tapered bottom surface that provides a level top resting surface for a transmission belt. It is recommended that each leveling loop is made out of a low friction flexible material that can expand and contract accordingly with the expansion and contraction of its cone; otherwise the CVT needs to be configured so that the leveling loops do not get in the way as the transmission ratio of their CVT is changed.

[0012] Tensioning Pulley **3** acts as a support pulley and tensioning pulley. Tensioning Pulley **3** and Support Pulley **4** are mounted so that they can freely move sideways in the horizontal direction. And Tensioning Pulley **3** is pushed upwards in the vertical direction so that it can maintain proper tension in Transmission Belt **2** for all operating conditions of the CVT.

[0013] Also, the “Mover Sliding Plate Mechanism” described in the “Mover Sliding Plate Mechanism for Converting Fixed Interval Movements to Required Interval Movement for Moving a Cone” has been previously disclosed in U.S. Pat. No. 9,958,064.

LISTING OF PRIOR ARTS

[0014] U.S. Pat. No. 7,713,153; Issue Date: 11 May 2010; Patentee: Naude. [0015] U.S. Pat. No. 6,656,070; Issue Date: 2 Dec. 2003; Patentee: Tay. [0016] U.S. Pat. No. 7,722,490; Issue Date: 25 May 2010; Patentee: Tay. [0017] U.S. Pat. No. 8,628,439; Issue Date: 14 Jan. 2014; Patentee: Tay. [0018] U.S. Pat. No. 9,651,123; Issue Date: 16 May 2017; Patentee: Tay. [0019] U.S. Pat. No. 9,958,064; Issue Date: 1 May 2018; Patentee: Tay. [0020] U.S. Pat. No. 9,964,209; Issue Date: 8 May 2018; Patentee: Tay. [0021] European Patent EP2764279; Issue Date: 17 Jan. 2018; Patentee: Tay. [0022] European Patent EP1311777; Issue Date: 6 Oct. 2010; Patentee: Tay.

BRIEF SUMMARY OF THE INVENTION

[0023] This disclosure describes a CVT that uses a “Cone with one Torque Transmitting Member” that is coupled by a Transmission Belt to another “Cone with one Torque Transmitting Member” for which teeth re-engagement of the “teeth of the Torque Transmitting Members” with the “teeth of their Transmission Belt” is improved.

[0024] For the CVT of prior art as described in the “BACKGROUND—DESCRIPTION OF

PRIOR ART” section, when used for heavy load applications (obviously the CVT will also be used under no load situations, such as during start-up), smooth tooth re-engagement can be problem because of the large “Transmission Belt Teeth Compression” and “Transmission Belt Stretching” variation on the tense-side of the Transmission Belt. Regarding “Transmission Belt Teeth Compression”, a heavy load will result in a large tension in the tense side of the Transmission Belt; and the large tension will push the teeth of said Transmission Belt towards its Cone at a force sufficient enough to cause significant compression of said teeth. Significant compression of said teeth will reduce the Transmission Diameter of the Transmission Belt, and this will cause the teeth of the Transmission Belt to slide circumferentially (away from their ideal engagement position) relative to the surface of their Cone; this is because the circumferential length of a portion of a Transmission Belt remains constant or almost constant regardless of its Transmission Diameter. The CVT's of this disclosure uses a Transmission Diameter Compensating Mechanism that can be used to address the “Transmission Belt Teeth Compression” and “Transmission Belt Stretching” problems in the tense-side of the Transmission Belt as discussed above.

[0025] In addition, for the CVT of prior art, when the axial positions of the Cones are changed independently as according to the method described in European Patent EP2764279; then slack needs to be provided or removed depending on whether the axial position of the Cones are “aligned” or “not aligned” for their current Transmission Ratio. And this can cause the tension in the slack side of the Transmission Belt to vary significantly between different Transmission Ratios; since the angles at which the Transmission Belt approaches and leaves its Tensioning Pulley can vary significantly between when slack needs to be provided and when slack needs to be removed.

[0026] The CVT's of this disclosure use a slack-side Tensioning Pulley in order to maintain the tension in the slack-side of the Transmission Belt almost constant so as to ensure smooth re-engagement of the Driven Cone; and a Transmission Diameter Compensating Mechanism, which increases the Transmission Diameter of the Driving Cone when the torque being pulled by the Driving Cone is increased, in order to compensate for an increase in “Transmission Belt Teeth Compression” and an increase in “stretching of the tense side Transmission Belt” due to an increase in tension in the tense side of the Transmission Belt when an increased amount of torque is being transmitted; to ensure smooth re-engagement of the Driving Cone.

[0027] The CVT's of this disclosure preferably also use a “Tensioning and Constraining Mechanism” that has a “slack side Constrainer” that removes or provides slack as needed when the axial position of the Cones are “aligned and “not aligned” for their current Transmission Ratio without substantially changing the “angles at which the Transmission Belt approaches and leaves its Tensioning Pulley”; so as to reduce/minimize changes of the “angles at which the Transmission Belt approaches and leaves its Tensioning Pulley” due different Transmission Ratios, and/or due to other operational variations of the CVT. Note, the Transmission Ratio depends on the axial position of the Cones, which can be “aligned” or “not aligned”.

[0028] And in order to compensate for: a) variation in “Transmission Belt Teeth Compression” at different Transmission Diameters; and b) variation in “slack side transmission belt tension” due to variation in the angle the Transmission Belt approaches and leaves its Tensioning Pulley based on whether the Cones are “aligned” or “not aligned” for their current Transmission Ratio, the CVT's of this disclosure preferably also use a “Method of changing the Transmission Diameters of the Cones in increments specifically selected for each Transmission Diameter and Transmission Ratio”.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING

[0029] FIG. 1 shows a partial front-view of a CVT that is constructed of out of two “cone with one torque transmitting member”. In FIG. 1, the cones are cut (so as to remove the surfaces in front of

said cut) at an axial-line selected so as to show the current torque transmitting member and non-torque transmitting member resting surfaces of the cones (refer to line A-A). In addition, all other surfaces of the cones, besides the current torque transmitting member and non-torque transmitting member resting surfaces of the cones, are not shown.

[0030] FIG. 2 shows a partial top-view of a CVT that is constructed of out of two “cone with one torque transmitting member”; note: some hidden lines are not shown.

[0031] FIG. 3 shows another partial front-view of a CVT that is constructed of out of two “cone with one torque transmitting member”. Like FIG. 1, in FIG. 3 the cones are cut (so as to remove the surfaces in front of said cut) at an axial-line selected so as to show the current torque transmitting member and non-torque transmitting member resting surfaces of the cones (refer to line B-B). In addition, all other surfaces of the cones, besides the current torque transmitting member and non-torque transmitting member resting surfaces of the cones, are not shown.

[0032] FIG. 4 shows another partial top-view of a CVT that is constructed of out of two “cone with one torque transmitting member”; note: some hidden lines are not shown.

[0033] FIG. 5 shows a front-view of a CVT that is constructed of out of two “cone with one torque transmitting member”, which uses a Tensioning Pulley 8 on the slack side of its Transmission Belt 7.

[0034] FIG. 6 shows a front-view of Transmission Diameter Compensating Mechanism 10 (Note: some hidden lines are not shown for clarity).

[0035] FIG. 7 shows a partial right-end-view of Transmission Diameter Compensating Mechanism 10.

[0036] FIG. 8 shows a front-view of Sleeve 11.

[0037] FIG. 9 shows a sectional-view of Sleeve 11.

[0038] FIG. 10 shows another sectional-view of Sleeve 11.

[0039] FIG. 11 shows another sectional-view of Sleeve 11.

[0040] FIG. 12 shows a front-view of Torsion Spring 13.

[0041] FIG. 13 shows a partial left-end-view of Torsion Spring 13.

[0042] FIG. 14 shows a partial right-end-view of Torsion Spring 13.

[0043] FIG. 15 shows a front-view of Nut Base 14.

[0044] FIG. 16 shows a right-end-view of Nut Base 14.

[0045] FIG. 17 shows a front-view of Nut 12.

[0046] FIG. 18 shows a right-end-view of Nut 12.

[0047] FIG. 19 shows a left-end-view of the Left-End Cover of Driving Cone 15.

[0048] FIG. 20 shows a front-view of Transmission Diameter Compensating Mechanism 10A (Note: some hidden lines are not shown for clarity).

[0049] FIG. 21 shows a front-view of Sleeve 11A.

[0050] FIG. 22 shows a partial right-end-view of Sleeve 11A.

[0051] FIG. 23 shows a front-view of Follower Mechanism Base 14A.

[0052] FIG. 24 shows a right-end-view of Follower Mechanism Base 14A.

[0053] FIG. 25 shows a left-end-view of Follower Mechanism 18.

[0054] FIG. 26 shows a front-view of Follower Mechanism 18.

[0055] FIG. 27 shows a left-end-view of Follower Frame 18-M1.

[0056] FIG. 28 shows a front-view of Follower Frame 18-M1.

[0057] FIG. 29 shows a left-end-view of Follower 18-M2.

[0058] FIG. 30 shows a front-view of Follower 18-M2.

[0059] FIG. 31 shows a left-end-view of Follower Mechanism 18A.

[0060] FIG. 32 shows a front-view of Transmission Diameter Compensating Mechanism 19.

[0061] FIG. 33 shows a front-view of Transmission Diameter Compensating Mechanism 19 for which the front-halves of Sleeve 20, Torsion Plate 27, and Locking Ring 28 have been removed by cutting them in half.

[0062] FIG. **34** shows a front-view of Sleeve **20**.
[0063] FIG. **35** shows a right-end-view of Sleeve **20**, for which some hidden-lines are not shown.
[0064] FIG. **36** shows a right-end-view of Sleeve **20**.
[0065] FIG. **37** shows a front-view of Torsion Spring **26**.
[0066] FIG. **38** shows a partial left-end-view of Torsion Spring **26**.
[0067] FIG. **39** shows a partial right-end-view of Torsion Spring **26**.
[0068] FIG. **40** shows a front-view of Stepper Motor **22**.
[0069] FIG. **41** shows a right-end-view of Stepper Motor **22**.
[0070] FIG. **42** shows a front-view of Spiral **23**.
[0071] FIG. **43** shows a right-end-view of Spiral **23**.
[0072] FIG. **44** shows a sectional-view of Spiral **23**.
[0073] FIG. **45** shows a front-view of Spiral **23**, for which the hidden-lines are not shown.
[0074] FIG. **46** shows another sectional-view of Spiral **23**.
[0075] FIG. **47** shows a back-view of Spiral **23**, for which the hidden-lines are not shown.
[0076] FIG. **48** shows a front-view of Torsion Plate **27**.
[0077] FIG. **49** shows a right-end-view of Torsion Plate **27**.
[0078] FIG. **50** shows a partial-front-view of Torsion Plate **27** when Torsion Plate **27** is oriented as shown in FIG. **49**.
[0079] FIG. **51** shows a front-view of Follower Mechanism **24**.
[0080] FIG. **52** shows a left-end-view of Follower Mechanism **24**.
[0081] FIG. **53** shows a front-view of Follower Mechanism **24** when Follower Mechanism **24** is oriented as shown in FIG. **52**.
[0082] FIG. **54** shows a front-view of Cone **25**.
[0083] FIG. **55** shows a left-end-view of Cone **25**, for which the hidden-lines are not shown.
[0084] FIG. **56** shows a right-end-view of Cone **25**, for which only Front Cover **25-S1** is shown.
[0085] FIGS. **57** and **58** show a partial side-view of a “Mover Sliding Plate Mechanism”.
[0086] FIG. **59** shows a partial end-view of a “Mover Sliding Plate Mechanism”.
[0087] FIG. **60** shows a top-view of a “Mover Sliding Plate Mechanism”.
[0088] FIG. **61** shows as a side-view of Mover Rod **36**.
[0089] FIG. **62** shows as a top-view of Mover Rod **36**.
[0090] FIG. **63** shows as a front-view of Mover Rod **36**.
[0091] FIG. **64** shows a partial front-view of a CVT that is constructed of out of two “cone with one torque transmitting member”, which uses the “Transmission Belt Constraining Method”. Similar to FIGS. **1** and **3**, in FIG. **64** the cones are cut (so as to remove the surfaces in front of said cut) at an axial-line selected so as to show the current torque transmitting member resting surfaces of the cones. In addition, all other surfaces of the cones, besides the current torque transmitting member resting surfaces of the cones, are not shown.
[0092] FIG. **65** shows another partial front-view of a CVT that is constructed of out of two “cone with one torque transmitting member”, which uses the “Transmission Belt Constraining Method”. The CVT shown in FIG. **65** is at a different transmission ratio than the CVT shown in FIG. **64**. Similar to FIGS. **1** and **3**, in FIG. **65** the cones are cut (so as to remove the surfaces in front of said cut) at an axial-line selected so as to show the current torque transmitting member resting surfaces of the cones. In addition, all other surfaces of the cones, besides the current torque transmitting member resting surfaces of the cones, are not shown.
[0093] FIG. **66** shows a front-view of “Tensioning and Constraining Mechanism **52**”.
[0094] FIG. **67** shows a side-view of “Tensioning and Constraining Mechanism **52**”.
[0095] FIG. **68** shows a front-view of Constrainer **53**.
[0096] FIG. **69** shows a side-view of Constrainer **53**.
[0097] FIG. **70** shows a top-view of Constrainer **53**.
[0098] FIG. **71** shows a sectional-view of Constrainer **53**.

[0099] FIG. **72** shows a front-view of Tensioning Pulley Slider Mechanism **54**.
[0100] FIG. **73** shows a side-view of Tensioning Pulley Slider Mechanism **54**.
[0101] FIG. **74** shows a top-view of Tensioning Pulley Slider Mechanism **54**.
[0102] FIG. **75** shows a front-view of Constrainer Push Plate **57**.
[0103] FIG. **76** shows a top-view of Constrainer Push Plate **57**.
[0104] FIG. **77** shows a sectional-view of Constrainer Push Plate **57**.
[0105] FIG. **78** shows a front-view of a Connecting Rod **58**.
[0106] FIG. **79** shows a front-view of a Frame **59-M1** of Constrainer Wheel Mounting Mechanism **45**.
[0107] FIG. **80** shows a side-view of a Frame **59-M1** of Constrainer Wheel Mounting Mechanism **45**.
[0108] FIG. **81** shows a top-view of a Frame **59-M1** of Constrainer Wheel Mounting Mechanism **45**.
[0109] FIG. **82** shows a front-view of a Constrainer Wheel **59-M2**, its Shaft **59-M3**, and its Slider Blocks **45-M4**.
[0110] FIG. **83** shows a side-view of a Constrainer Wheel **59-M2**, its Shaft **59-M3**, and its Slider Blocks **59-M4**.
[0111] FIG. **84** shows a front-view of “Tensioning and Constraining Mechanism **52A**”.
[0112] FIG. **85** shows a side-view of “Tensioning and Constraining Mechanism **52A**”.
[0113] FIG. **86** shows a partial front-view of a “CVT 4 of a CVT 6” that uses the “Transmission Belt Constraining Method”. Similar to FIGS. **1** and **3**, in FIG. **86** the cones are cut (so as to remove the surfaces in front of said cut) at an axial-line selected so as to show the current torque transmitting member resting surfaces of the cones. In addition, all other surfaces of the cones, besides the current torque transmitting member resting surfaces of the cones, are not shown.
[0114] FIG. **87** shows a front-view of “Tensioning and Constraining Mechanism **52B**”.
[0115] FIG. **88** shows a side-view of “Tensioning and Constraining Mechanism **52B**”.
[0116] FIG. **89** shows a front-view of a Connecting Rod **58A**.
[0117] FIG. **90** shows a front-view of Top Plate **65**.
[0118] FIG. **91** shows a side-view of Top Plate **65**.
[0119] FIG. **92** shows a top-view of Top Plate **65**.
[0120] FIG. **93** shows a front-view of a CVT that is constructed out of two “cone with two oppositely positioned torque transmitting members”, which uses a Tensioning Pulley **8** on the slack side of its Transmission Belt **7**.
[0121] FIG. **94** shows a partial front-view of a CVT that is constructed out of two “cone with two oppositely positioned torque transmitting members”, which uses the “Transmission Belt Constraining Method”. Similar to FIGS. **1** and **3**, in FIG. **94** the cones are cut (so as to remove the surfaces in front of said cut) at an axial-line selected so as to show the current torque transmitting members resting surfaces of the cones. In addition, all other surfaces of the cones, besides the current torque transmitting members resting surfaces of the cones, are not shown.

REFERENCE NUMERALS IN DRAWINGS

[0122] For the reference numerals in this disclosure, the label M (number) after a reference numeral, where (number) is a number, such as M2 for example, is used to label different members of a part that is given one reference numeral but consist of more than one member. And the label S (number) after a reference numeral, where (number) is a number, such as S2 for example, is used to label the different shapes of a part that is given one reference numeral. Furthermore, same parts that are used in different location might have a different labeling letter after their reference numeral, or a different reference numeral altogether if this is helpful in describing the invention. If two parts have the same reference numeral then they are identical unless otherwise described.

DETAILED DESCRIPTION OF THE INVENTION

CVT Configuration for usage with a “Transmission Diameter Compensating Mechanism” and/or

“Tensioning and Constraining Mechanism” (FIG. 5)

[0123] A CVT configuration that can be used with a “Transmission Diameter Compensating Mechanism” and/or “Tensioning and Constraining Mechanism” as described in this disclosure is shown in FIG. 5.

[0124] The CVT shown in FIG. 5 comprises of a Driving Cone 5 (which has a Torque Transmitting Member 5-M1) that is coupled by a Transmission Belt 7 to a Driven Cone 6 (which has a Torque Transmitting Member 6-M1); in a manner so that the apex of Driving Cone 5 and the apex of Driven Cone 6 point in opposite directions (see configuration shown in FIG. 2). In addition, said CVT also has a Support Pulley 9 and Tensioning Pulley 8, which is positioned on the slack side of Transmission Belt 7 and pushed-up by a spring or other means. Note, when the Tensioning Pulley is placed on the slack side, its pre-tension should be lower than when it is placed on the tense side, since the purpose of the tensioner is to maintain adequate slack side tension.

[0125] It is desired to minimize the variation in the Transmission Diameter of a Cone due to variation in Transmission Belt tension, so that tooth misalignment during re-engagement of a Torque Transmitting Member due to variation in Transmission Belt tension can be eliminated/minimized.

[0126] For the CVT shown in FIG. 5, with sufficient pre-tension, the tension in the slack side of the transmission belt should be nearly constant regardless of the torque transmitted by the CVT; so that no Transmission Diameter Compensating Mechanism 10 or 10A is required for Driven Cone 6, which engages on the slack side of the transmission belt.

[0127] For the CVT shown in FIG. 5, the tension in the tense side of the transmission belt depends of the torque transmitted by the CVT; so in order to minimize changes in the Transmission Diameter of the cone due to variation in tension in the tense side of the transmission belt, a Transmission Diameter Compensating Mechanism 10 or 10A is used for Driving Cone 5, which engages on the tense side of the transmission belt.

Transmission Diameter Compensating Mechanism (FIGS. 6 to 19)

[0128] A Transmission Diameter Compensating Mechanism 10 is a mechanism that pushes its Driving Cone 15 in the axial direction that will increase its Transmission Diameter when the torque being pulled by its Driving Cone 15 is increased; this is in order to compensate for an increase in “Transmission Belt Teeth Compression” and an increase in “stretching of the tense side Transmission Belt” due to an increase in tension in the tense side of its Transmission Belt.

[0129] Transmission Diameter Compensating Mechanism 10 is shown in FIGS. 6 and 7.

Transmission Diameter Compensating Mechanism 10 comprises of a Sleeve 11 that slides on a Spline 17. Sleeve 11 has a Threaded Section 11-S7. On Threaded Section 11-S7, a Nut 12 is screwed on. One end of a Torsion Spring 13 is rotatably and axially fixed to Sleeve 11; while the other end of Torsion Spring 13 is rotatably fixed to a Nut Base 14. Nut Base 14 is shaped so it can transfer the rotation of Torsion Spring 13 to Nut 12, while allowing Nut 12 to freely move axially relative to Torsion Spring 13. Nut 12 is fixed to its Driving Cone 15, so that Driving Cone 15 is axially and rotatably constrained relative to Nut 12.

[0130] Spline 17 is pulling torque in the direction shown by Arrow 16, the direction of the pulling-resistance/reaction-force of Driving Cone 15 is shown by Arrow 16A. Here when the torque transmitted by Driving Cone 15 is increased (which increases the reaction-force of Driving Cone 15 applied to Nut 12), Torsion Spring 13 will deflect more so as to allow Nut 12 to rotate relative to Threaded Section 11-S7, which then (because of the threaded connection) will move Driving Cone 15 in the direction that will increase its Transmission Diameter.

[0131] Sleeve 11 is shown in FIGS. 8 to 11. In order to fix the bottom-end of a Torsion Spring 13 to Sleeve 11, as required per the previous paragraph, a Spring Connecting Plate 11-S4 is fixed to Sleeve 11 (see FIGS. 8 and 10). Spring Connecting Plate 11-S4 has a Hole 11-S5 into which the Insertion Rod of Mounting Shape 13-S1 of Torsion Spring 13 is inserted during assembly.

[0132] In order to prevent Nut 12 and Nut Base 14 from moving in the direction that will decrease

the Transmission Diameter of Driving Cone **15** (which is to the left), two Stops **11-S6** are fixed to Sleeve **11**. The engagement of Stops **11-S6** with Nut Base **14** will prevent Nut **12**, Nut Base **14**, and Driving Cone **15** from moving to the left in instances where the direction of the pulling torque is reversed.

[0133] In order to connect Sleeve **11** to the mechanism used to change the axial position of its Driving Cone **15**, Sleeve **11** has a Flange **11** and Locking Ring Groove **11-S2**. Here a Bearing of a Connecting Rod of a Transmission Ratio Changing Mechanism can be inserted into Sleeve **11** and then axial secured through its engagement with Flange **11** and the Locking Ring inserted into Locking Ring Groove **11-S2**.

[0134] Torsion Spring **13** is shown in FIGS. **12** to **14**. In order to attach the bottom-end of Torsion Spring **13** to Spring Connecting Plate **11-S4** of Sleeve **11**, the bottom-end of Torsion Spring **13** has a Mounting Shape **13-S1**. Mounting Shape **13-S1** has a Collar (which engages with a side surface of Spring Connecting Plate **11-S4**), an Insertion Rod (which will be inserted into Hole **11-S5** of Spring Connecting Plate **11-S4**), and a Locking Pin Hole (which will be used to secure the bottom-end of Torsion Spring **13** to Spring Connecting Plate **11-S4** through the use of a Locking Pin, which is not shown).

[0135] In order to attach the top-end of Torsion Spring **13** to Spring Connecting Plate **14-S2** of Nut Base **14**, the top-end of Torsion Spring **13** has a Mounting Shape **13-S2**. Mounting Shape **13-S2** has a Collar (which engages with the one side surface of Spring Connecting Plate **14-S2**), an Insertion Rod (which will be inserted into Hole **14-S3** of Spring Connecting Plate **14-S2**), and a Locking Pin Hole (which will be used to secure the bottom-end of Torsion Spring **13** to Spring Connecting Plate **14-S2** through the use of a Locking Pin, which is not shown).

[0136] Nut Base **14** is shown in FIGS. **15** and **16**. It is shaped like a disc that has a Hole **14-S1**, which is used to insert Nut Base **14** through the front-end of Slide **11**. Hole **14-S1** has a diameter that is large enough relative to the diameter of Sleeve **11** so that it can freely slide relative to Sleeve **11**.

[0137] In order to fix the top-end of a Torsion Spring **13** to Nut Base **14**, a Spring Connecting Plate **14-S2** is fixed to Nut Base **14**. Spring Connecting Plate **14-S2** has a Hole **14-S3** into which the Insertion Rod of Mounting Shape **13-S2** of Torsion Spring **13** is inserted during assembly.

[0138] In order to allow Nut Base **14** to transfer the rotation of Torsion Spring **13** to Nut **12**, while allowing Nut **12** to freely move axially (to the left or to the right) relative to Torsion Spring **13** (as required by the first paragraph of this section), Nut Base **14** has two Rods **14-S4**; which during assembly are inserted into Rod Holes **12-S2** of Nut **12**, so as to allow Nut **12** to freely move axially relative to Nut Base **14**. Here the engagement of Rods **14-S4** with Rod Holes **12-S2** will transfer the rotation of Nut Base **14** (which is fixed for rotation relative to Torsion Spring **13**) to Nut **12**.

[0139] Rods **14-S4** are long enough so that they extend beyond Nut **12** and also engage with Holes **15-S2** of the left-end cover of Driving Cone **15** (see FIG. **6**; and FIG. **19**, which shows a left-end-view of the Left-End Cover of Driving Cone **15**); this will rotatably constrain Nut **12** relative to Driving Cone **15** (as required by the first paragraph of this section).

[0140] Driving Cone **15** should be allowed to freely slide relative to Spline **17** and Sleeve **11**; for this purpose, the left-end cover of Driving Cone **15** has a Hole **15-S1** for Sleeve **11** (see FIG. **19**), and the right-end cover of Driving Cone **15** has a Spline Cut-out **15-S3** for Spline **17** (see FIG. **7**). Since Driving Cone **15** can slide freely relative to Sleeve **11**, it is forced towards Nut **12** by the tension of its Transmission Belt, so as to axially constrain Driving Cone **15** relative to Nut **12** (as required by the first paragraph of this section). If desired, Driving Cone **15** can also be axially and rotatably constrained relative to Nut **12** through gluing, screwed connections, etc.

[0141] Nut **12** is shown in FIGS. **17** and **18**. It is shaped like a round nut that has a centric Threaded Hole **12-S1** and two Rod Holes **12-S2**. When Transmission Diameter Compensating Mechanism **10** is assembled (see FIG. **6**), Threaded Hole **12-S1** is screwed on to Threaded Section **11-S7** of Sleeve **11**. Here because of the threaded engagement, rotating Nut **12** counter-clockwise

(which can be caused by the increased flexing of Torsion Spring **13** due to the increased torque applied on Driving Cone **15**) will move Nut **12** to the right so as to increase the Transmission Diameter of Driving Cone **15**.

Transmission Diameter Compensating Mechanism with Spiral

[0142] Transmission Diameter Compensating Mechanism **10A**, shown in FIG. **20**, is identical to Transmission Diameter Compensating Mechanism **10**, except that the “Screw-and-Nut Mechanism” of Transmission Diameter Compensating Mechanism **10** is replaced with a “Spiral-and-Follower Mechanism”. Because of this, the following parts of Transmission Diameter Compensating Mechanism **10** are replaced in Transmission Diameter Compensating Mechanism **10A**: Sleeve **11** is replaced with Sleeve **11A**, Nut Base **14** is replaced with Follower Mechanism Base **14A**, and Nut **12** is replaced with Follower Mechanism **18**.

[0143] Sleeve **11A** is shown in FIGS. **21** and **22**. It is identical to Sleeve **11**; except that here, Threaded Section **11-S7** is replaced with a Spiral Groove **11A-S7**. Spiral Groove **11A-S7** is formed in-between a Back Spiral Wall **11A-S8** and a Front Spiral Wall **11A-S9**, which are parallel to each other (see FIG. **21**).

[0144] Follower Mechanism Base **14A** is shown in FIGS. **23** and **24**. It is identical to Nut Base **14**; except that for Follower Mechanism Base **14A**, the two “Rods **14-S4** of Nut Base **14**” are replaced with “Rods **14A-S4**”. Rods **14A-S4** are identical to Rods **14-S4** except for being longer. This is required in order to ensure that Rods **14A-S4** are always engaged with Holes **18-M1-S2** of Follower Mechanism **18** (see FIGS. **27** and **28**), and are also always engaged with Holes **15-S2** of the left-end cover of Driving Cone **15** (see FIG. **19**).

[0145] Follower Mechanism **18** is shown in FIGS. **25** to **30**. Follower Mechanism **18** has the same function and operates identically as Nut **12**; except that for Follower Mechanism **18**, the threads of Nut **12** that engage with Threaded Section **11-S7** of Sleeve **11**, are replaced with a Follower **18-M2** that engages with Spiral Groove **11A-S7** of Sleeve **11A**.

[0146] Follower Mechanism **18** comprises of a Follower Frame **18-M1** (see FIGS. **27** and **28**), a Follower **18-M2** (see FIGS. **29** and **30**), and Locking Ring **18-M3** (see FIGS. **25** and **26**). Follower **18-M2** is inserted into Hole **18-M1-S3** of Follower Frame **18-M1**; in a manner so that it can rotate relative to Follower Frame **18-M1**, but is constrained for any other movements relative to Follower Frame **18-M1**. And Locking Ring **18-M3** is used to secure Follower **18-M2** to Follower Frame **18-M1**.

[0147] Follower Frame **18-M1** is free to slide axially, but is rotationally constrained relative to Follower Mechanism Base **14A**; and the engagement of Rods **14A-S4** of Follower Mechanism Base **14A** with the Rod Holes **18-M1-S2** of Follower Frame **18-M1** will also ensure that Follower Frame **18-M1** is always parallel relative Follower Mechanism Base **14A**. Because of this set-up, Follower Frame **18-M1** can be used to properly constrain the position of Follower **18-M2** relative Spiral Groove **11A-S7** in a manner so that the engagement of Follower **18-M2** with Spiral Groove **11A-S7** can be used to accurately control the axial position of Follower Frame **18-M1**, and such also the axial position of Driving Cone **15**.

[0148] Follower Frame **18-M1** is shown in FIGS. **27** and **28**. It is shaped like a disc that has a centric Sleeve Hole **18-M1-S1**, which allows Follower Frame **18-M1** to freely slide axially relative to Sleeve **11A**; and two Rod Holes **18-M1-S2**, into which Rods **14A-S4** of Follower Mechanism Base **14A** are inserted in a manner so as to allow Rods **14A-S4** to slide axially relative to Rod Holes **18-M1-S2** while restricting any other relative movements between Rods **14A-S4** and Rod Holes **18-M1-S2**.

[0149] And in order to attach Follower **18-M2** to Follower Frame **18-M1**, Follower Frame **18-M1** has an Attachment Plate **18-M1-S3**, which is a plate that has a hole into which the Shaft **18-M2-S1** of Follower **18-M2** is inserted. The fit between the hole of Attachment Plate **18-M1-S3** and Shaft **18-M2-S1** should be loose enough to allow Shaft **18-M2-S1** to freely rotate relative to Attachment Plate **18-M1-S3**, but tight enough to restrict any radial movements between Shaft **18-M2-S1** and

Attachment Plate **18-M1-S3**.

[0150] Follower **18-M2** is shown in FIGS. **29** and **30**. The main shape of Follower **18-M2** is Shaft **18-M2-S1**. At the left-end of Shaft **18-M2-S1**, two Back Engaging Surfaces **18-M2-S2** are shaped. Back Engaging Surfaces **18-M2-S2** can be shaped by separate shapes that are fixed to Shaft **18-M2-S1** (such as by gluing, bonding, etc.), or by machining the shapes onto Shaft **18-M2-S1**. Back Engaging Surfaces **18-M2-S2** are the surfaces of Follower **18-M2** that engage with Back Spiral Wall **11A-S8** of Spiral Groove **11A-S7**. Here in order to reduce the Hertzian Contact Stresses, Back Engaging Surfaces **18-M2-S2** should have a partial cylindrical contact surface, which has the largest diameter that allows for smooth engagement with Back Spiral Wall **11A-S8** of Spiral Groove **11A-S7**. It is recommended that Back Engaging Surfaces **18-M2-S2** are shaped to the sides of the point of contact of Shaft **18-M2-S1** (see FIG. **30**), so as to “not increase the diameter of said point of contact”; so that Shaft **18-M2-S1** can smoothly rotate as needed to have Back Engaging Surfaces **18-M2-S2** maintain normal engagement with Back Spiral Wall **11A-S8** as the shape of Back Spiral Wall **11A-S8** changes.

[0151] Shaft **18-M2-S1** does not have any “Hertzian Stress Reducing Engaging Surfaces” for its engagement with Front Spiral Wall **11A-S9**. So that the cylindrical surface of Shaft **18-M2-S1** will engage directly with Front Spiral Wall **11A-S9**. The forces on Front Spiral Wall **11A-S9** due to its engagement with Shaft **18-M2-S1** are only due to the force needed to change the axial position of its cone in the direction that that pulls its cone away from the axial compression applied on its cone by the tension of its transmission belt; these forces are lower than the force applied on Back Spiral Wall **11A-S8**, which has to resist the axial compression applied on its cone by the tension of its transmission belt.

[0152] At the right-end of Shaft **18-M2-S1**, a Locking Ring Groove **18-M2-S3** is shaped (see FIG. **29**). During assembly, the right-end of Shaft **18-M2-S1** is inserted into the hole of Attachment Plate **18-M1-S3**; and then Shaft **18-M2-S1** is secured to Attachment Plate **18-M1-S3** by inserting a Locking Ring **18-M3** into Locking Ring Groove **18-M2-S3**.

[0153] An alternate to Follower Mechanism **18** is shown in FIG. **31**, where it is labeled as Follower Mechanism **18A**. It is identical to Follower Mechanism **18**, except that its Follower Frame **18A-M1** has a Counterbalance **18A-M1-S4**. Counterbalance **18A-M1-S4** is used to counterbalance the masses of Attachment Plate **18-M1-S3**, Follower **18-M2**, and the Locking Ring that is used to secure Follower **18-M2** to Attachment Plate **18-M1-S3**.

[0154] If desired, Transmission Diameter Compensating Mechanisms **10** and **10A** can be replaced with a sensor that monitors and an actuator that controls the actual Transmission Diameter of a Cone. Transmission Diameter Compensating Mechanism **76** can also be used with a Cone that engages on the slack side of its transmission belt; but here depending on the configuration of the CVT, the threads might be angled in the opposite direction. For example, for a CVT where the tensioning pulley is positioned on the tense side, then if used, the Transmission Diameter Compensating Mechanism **76** should move its cone that engages on the slack of its transmission belt in the direction that will decrease its Transmission Diameter as the torque is increased, since here an increase in torque will reduce slack side tension and as such compression.

Transmission Diameter Compensating Mechanism with Adjustable Spiral
Overview

[0155] A Transmission Diameter Compensating Mechanism **19**, which can be used instead of Transmission Diameter Compensating Mechanism **10** is shown in FIG. **32**. For Transmission Diameter Compensating Mechanism **19**, the “Screw-and-Nut Mechanism” of Transmission Diameter Compensating Mechanism **10** is replaced with a “Spiral-and-Follower Mechanism”.

[0156] Like Transmission Diameter Compensating Mechanism **10**, Transmission Diameter Compensating Mechanism **19** should be used for a Driving Cone, and the spline of Transmission Diameter Compensating Mechanism **19** should pull torque in the same direction as the spline of Transmission Diameter Compensating Mechanism **10**.

[0157] The “Spiral-and-Follower Mechanism” of Transmission Diameter Compensating Mechanism **19** can be used to provide more precise “Cone Diameter Increase Compensation”; since it allows for linear and non-linear compensation profiles, instead of just a linear compensation profile as can be provided by the “Screw-and-Nut Mechanism” of Transmission Diameter Compensating Mechanism **10**.

[0158] Furthermore, the profile of the Spiral of the “Spiral-and-Follower Mechanism” of Transmission Diameter Compensating Mechanism **19** can be adjusted to an ideal profile for the “current Transmission Diameter of its Cone” by rotating said Spiral, preferably before the Transmission Diameter of its Cone is changed for a “CVT for which the tension in its Transmission Belt can be reduced before and during Transmission Diameter Changing”.

[0159] Changing the profile of the Spiral can provide better “Cone Diameter Increase Compensation” because for a “Small Transmission Diameter” of a Cone, the compression force of its Transmission Belt is dispersed over a smaller surface area compared to a “Large Transmission Diameter” of said Cone; so that for a “Small Transmission Diameter” of a Cone more “Cone Diameter Increase Compensation” needs to be provided to compensate for the compression force of its Transmission Belt compared to a “Large Transmission Diameter” of said Cone.

[0160] And changing the profile of the Spiral can also adjust the “Cone Diameter Increase Compensation Provided” for compensating for stretching of its Transmission Belt. Here changing the profile of the Spiral can compensate for variation in stretching of its Transmission Belt at different Transmission Diameters due to length variation of said Transmission Belt at different Transmission Diameters.

[0161] Transmission Diameter Compensating Mechanism **19**, shown in FIGS. **32** and **33**, is basically identical to Transmission Diameter Compensating Mechanism **10**, except that the “Screw-and-Nut Mechanism” of Transmission Diameter Compensating Mechanism **10** is replaced with an adjustable “Spiral-and-Follower Mechanism”. This modification changes the parts of Transmission Diameter Compensating Mechanism **17** as described below.

[0162] Transmission Diameter Compensating Mechanism **19** has a Sleeve **20**, which has a centric Spline Profile Hole **20-S1** (see FIGS. **34** and **35**). A Spline **21** is inserted Into Spline Profile Hole **20-S1** in a manner so that Sleeve **20** cannot rotate relative to Spline **21**, but can slide axially relative Spline **21**.

[0163] Sleeve **20**, which is shown in detail in FIGS. **34** to **36**, is basically identical to Sleeve **11** of Transmission Diameter Compensating Mechanism **10**, except that the Threaded Section **11-S7** of Sleeve **11** is replaced with a Cavity **20-S2** for mounting a Stepper Motor **22** to which a Spiral **23** is mounted (see FIGS. **33** and **34**). Transmission Diameter Compensating Mechanism **19** is used for a Cone **25**.

[0164] For Transmission Diameter Compensating Mechanism **19** the “rotational-deflection to axial movement conversion of Cone **25**” is provided by the interaction between Spiral **23** and Followers **24-S5**, which are fixed to a Follower Disk **24-S1** of a Follower Mechanism **24** (see FIGS. **32**, and **51** to **53**). During torque transmission operation; Spiral **23** is rotationally-fixed relative to Sleeve **20**, while Follower Disk **24-S1**, and as such Followers **24-S5**, rotate with their Cone **25** relative to Sleeve **20** based on the amount of rotational deflection of a Torsion Spring **26** due to the torque transmitted by Cone **25**.

[0165] Spiral **23** is shown in FIGS. **42** to **47**; Drawings Set 1: FIG. **42** shows a front-view of Spiral **23** and FIG. **43** shows a right-end-view of Spiral **23**; Drawings Set 2: FIG. **45** shows a front-view of Spiral **23** (which is the left-view of Spiral **23** when Spiral **23** is oriented as shown in FIG. **46**), FIG. **47** shows a back-view of Spiral **23** (which is the right-view of Spiral **23** when Spiral **23** is oriented as shown in FIG. **46**), and FIGS. **44** and **46** show sectional-views of Spiral **23**. The hidden lines are not shown in FIGS. **44** to **47**.

[0166] Spiral **23** is shaped like cylinder that has a Front Outer Spiral Groove **23-S2** and Front Inner Spiral Groove **23-S4** shaped on its front-cylindrical-half (see FIG. **45**); and a Back Outer Spiral

Groove **23-S6** and Back Inner Spiral Groove **23-S8** shaped on its back-cylindrical-half (see FIG. **47**).

[0167] During assembly, Follower Mechanism **24** (see FIGS. **51** to **53**) is threaded onto Spiral **23** by first inserting a Follower **24-S5** into each, Follower Insertion Slot **23-S3** and Follower Insertion Slot **23-S7** (see FIGS. **45** to **47**); and then rotating Follower Mechanism **24** so as to guide one Follower **24-S5** to its final assembled position in Front Outer Spiral Groove **23-S2**, and so as to also guide the other Follower **24-S5** to its final assembled position in Back Outer Spiral Groove **23-S6**.

[0168] The profile of Front Outer Spiral Groove **23-S2** mirrors the profile of Back Outer Spiral Groove **23-S6** in a manner such that the rotation of Follower Mechanism **24** relative to Spiral **23** will cause “the axial movement applied to Follower Mechanism **24** due to the engagement of Front Outer Spiral Groove **23-S2** with its Follower **24-S5**” to be the same as “the axial movement applied to Follower Mechanism **24** due to the engagement of Back Outer Spiral Groove **23-S6** with its Follower **24-S5**”; so that Follower Mechanism **24** moves axially relative to Sleeve **20** in a manner so that Follower Disk **24-S1** is always parallel relative to the right-end surface of Sleeve **20**.

[0169] In order to be able rotate Spiral **23** and control the axial position of Spiral **23**, without having Front Outer Spiral Groove **23-S2** and Back Outer Spiral Groove **23-S6** apply any axial-forces on their Followers **24-S5**, when their Followers **24-S5** are at their neutral position; Spiral **23** also has a Front Inner Spiral Groove **23-S4** (see FIG. **45**) and Back Inner Spiral Groove **23-S8** (see FIG. **47**).

[0170] During assembly, Spiral **23** is threaded onto Sleeve **20** by first inserting a Spiral Positioning Pin **20-S10** (see FIGS. **34** and **35**) into each, Pin Insertion Slot **23-S5** and Pin Insertion Slot **23-S9** (see FIGS. **44**, **45**, and **47**); and then rotating Spiral **23** so as to guide one Spiral Positioning Pin **20-S10** to its final assembled position in Front Inner Spiral Groove **23-S4**, and so as to also guide the other Spiral Positioning Pin **20-S10** to its final assembled position in Back Inner Spiral Groove **23-S8**.

[0171] The engagement of the Spiral Positioning Pins **20-S10** with their Inner Spiral Grooves (Front Inner Spiral Groove **23-S4** or Back Inner Spiral Groove **23-S8** respectively), is used to control the axial position of Spiral **23** relative to Sleeve **20**.

[0172] Front Inner Spiral Groove **23-S4** is “parallel to” and has the same profile as Front Outer Spiral Groove **23-S2**; and Back Inner Spiral Groove **23-S8** is “parallel to” and has the same profile as Back Outer Spiral Groove **23-S6**; so that when Followers **24-S5** are at their neutral position, which is the position where they are horizontally aligned with Positioning Pins **20-S10**, the axial distance between “the Spiral Positioning Pins **20-S10** positioned in the Inner Spiral Grooves (Front Inner Spiral Groove **23-S4** and Back Inner Spiral Groove **23-S8**)” and “the Followers **24-S5** positioned in the Outer Spiral Grooves (Front Outer Spiral Groove **23-S2** and Back Outer Spiral Groove **23-S6**)” remains constant regardless of the rotational position of Spiral **23**; so that when Followers **24-S5** are at their neutral position, no axial force is applied to Followers **24-S5** due to the rotation of Spiral **23**.

[0173] Preferably, the “Spiral Grooves Profiles” of Spiral **23** (which are Front Outer Spiral Groove **23-S2**, Front Inner Spiral Groove **23-S4**, Back Outer Spiral Groove **23-S6**, and Back Inner Spiral Groove **23-S8**) have a lead angle that is small enough so as to prevent any axial forces applied on them by Followers **24-S5** and by Spiral Positioning Pins **20-S10** from rotating their Spiral **23**.

[0174] It is also recommended that Spiral **23** is at its most retracted position, which is a position where it cannot be rotated into its Sleeve **20** any further, when its CVT is shut-down. This is to prevent accidental/unintended rotation of Spiral **23** when its Stepper Motor **22** is turned-off, since Stepper Motor **22** has preferably some torque that prevents accidental/unintended rotation of Spiral **23** when it is turned-on.

[0175] If the Spirals **23** of a CVT are at their most retracted positions when their CVT are Shut-down, then the “Spiral Grooves Profiles” of the Cone(s) which is/are at its/their smallest

Transmission Diameter when its/their CVT is/are Shut-down should differ from the “Spiral Grooves Profiles” of the Cone(s) which is/are at its/their largest Transmission Diameter when its/their CVT is/are Shut-down. The “Spiral Grooves Profiles” of a Spiral **23** can be obtained through experimentation or fundamental mathematics/engineering.

[0176] In order to rotate Spiral **23** with a hexagonal-shaped Shaft **22-M1** of Stepper Motor **22**, Spiral **23** has a hexagonal Hole **23-S1** for Shaft **22-M1** (see FIGS. **42** and **43**). The fit between Hole **23-S1** and Shaft **22-M1** is such that Spiral **23** can smoothly slide axially relative Shaft **22-M1**, so as to allow Spiral **23** to move axially relative Sleeve **20**; while minimizing relative rotational movement between Spiral **23** and Shaft **22-M1**. And in order to limit how much Spiral **23** can retract into its Sleeve **20**, Spiral **23** has a Cover **23-S10** at its front-end (see FIG. **42**). The engagement of the rear surface of Cover **23-S10** with the front-end of Shaft **22-M1** will limit how much Spiral **23** can retract into its Sleeve **20**.

[0177] The left-end of Torsion Spring **26** is fixed to Sleeve **20**, while the right-end of Torsion Spring **26** is fixed to Torsion Plate **27**. Torsion Plate **27** is mounted on Sleeve **20** so that it can rotate relative to Sleeve **20** but not move axially relative to Sleeve **20**; a Sleeve Shoulder **20-S7** and a Locking Ring **28** are used to constrain the axial position of Torsion Plate **27** relative to Sleeve **20** (see FIGS. **32** and **33**). And since Torsion Plate **27** is fixed to the right-end of Torsion Spring **26**, its rotation relative to Sleeve **20** will cause Torsion Spring **26** to flex/deflect.

[0178] Torsion Plate **27** (which is shown by itself in FIGS. **48** to **50**) has two Rods **27-S3**, which engage with Rod Holes **24-S3** of Follower Mechanism **24** (see FIGS. **51** and **52**) in a manner so as to transfer the rotation of Torsion Plate **27** to Follower Mechanism **24**, while allowing Follower Mechanism **24** to move axially relative to Torsion Plate **27**. And Cone **25** has a Rear Cover **25-S3** (see FIGS. **54** and **55**) that has two Rod Holes **25-S4** that also engage with Rods **27-S3** of Torsion Plate **27** in a manner so as to transfer the rotation of Cone **25** to Torsion Plate **27**, while allowing Cone **25** to move axially relative to Torsion Plate **27**.

[0179] Through the arraignment of the previous paragraphs, the amount of torque transmitted by Cone **25** will determine the amount of rotation of Torsion Plate **27** relative to Sleeve **20** (and as such also Spiral **23**) due to the deflection of Spring **26**. And because of the rotational-to-axial movement conversion due to the engagement of “Followers **24-S5** of Follower Mechanism **24**” with the “Outer Spiral Grooves of Spiral **23**”, the amount of rotation of Torsion Plate **27** relative to Spiral **23** will determine the amount of “Cone Diameter Increase Compensation” provided by Transmission Diameter Compensating Mechanism **19**.

Details of Individual Parts—Sleeve **20**

[0180] Sleeve **20** is shown in detail by itself in FIGS. **34** to **36**; Sleeve **20** with its assembled parts is shown in FIGS. **32** and **33**. Sleeve **20** has the following shapes: a) a Spline Profile Hole **20-S1** for a Spline **21**, b) a centric Cavity **20-S2** for mounting Stepper Motor **22**, c) Threaded Holes **20-S3** for fastening Stepper Motor **22** to Cavity **20-S2**, d) a Spring Connecting Plate **20-S4** for rotate-ably fixing the left end of Torsion Spring **26** to Sleeve **20**, e) a Connecting Rod Shaft **20-S6** for inserting the bearing of Connecting Rod **29**, which is used to change the axial position of Cone **25**, f) a Locking Ring Groove **20-S7** for a Locking Ring **30**, which is used to axially constrain Connecting Rod **29** to Sleeve **20**; e) two oppositely positioned Sleeve Shoulders **20-S8**; f) a Locking Ring Groove **20-S9** for a Locking Ring **28**, which with Sleeve Shoulders **20-S8** are used to axially constrain the axial position of Torsion Plate **27** relative to Sleeve **20** (see FIG. **33**); and g) two oppositely positioned Spiral Positioning Pins **20-S10**, which are fixed to Cavity **20-S2** as shown in FIGS. **34** to **36**.

[0181] Spring Connecting Plate **20-S4** (see FIGS. **34** to **35**) has a Hole **20-S5** for inserting a Mounting Rod **26-S1** of Torsion Spring **26** (see FIGS. **37** to **39**). Hole **20-S5** has the same dimension as Hole **27-S5** of Spring Connecting Plate **27-S4** of Torsion Plate **27**. Hole **27-S5**, which is shown at its opening in FIG. **50**, is used for inserting a Mounting Rod **26-S2**, which has the same dimension as Mounting Rod **26-S1**.

[0182] In order to allow for dimensional changes of Torsion Spring **26** due to flexing, Hole **20-S5** and Hole **27-S5** can be made slightly larger than the Mounting Rod inserted into each of them, which is either Mounting Rod **26-S1** or Mounting Rod **26-S2** respectively.

Details of Individual Parts—Torsion Spring **26**

[0183] Torsion Spring **26** is shown in FIGS. **37** to **39**. In order to attach the left-end of Torsion Spring **26** to Spring Connecting Plate **20-S4** of Sleeve **20**, the left-end of Torsion Spring **26** has a Mounting Rod **26-S1**. Mounting Rod **26-S1** has a Collar **26-S1-S1** (which engages with a side surface of Spring Connecting Plate **20-S4**, see FIGS. **34**, **38**, and **32**), an Insertion Rod **26-S1-S2** (which is inserted into Hole **20-S5** of Spring Connecting Plate **20-S4**, see FIGS. **34**, **38**, and **32**), and a Locking Pin Hole **26-S1-S3** (which is used to secure the left-end of Torsion Spring **26** to Spring Connecting Plate **20-S4** through the use of a Locking Pin **31**, see FIGS. **34**, **38**, and **32**).

[0184] In order to attach the right-end of Torsion Spring **26** to Spring Connecting Plate **27-S4** of Torsion Plate **27**, the right-end of Torsion Spring **26** has a Mounting Rod **26-S2**. Mounting Rod **26-S2** has a Collar **26-S2-S1** (which engages with a side surface of Spring Connecting Plate **27-S4**, see FIGS. **39**, **45**, and **32**), an Insertion Rod **26-S2-S2** (which will be inserted into Hole **27-S5** of Spring Connecting Plate **27-S4**, see FIGS. **34**, **45**, and **32**), and a Locking Pin Hole **26-S2-S3** (which will be used to secure the right-end of Torsion Spring **26** to Spring Connecting Plate **27-S4** through the use of another Locking Pin **31**, see FIGS. **34**, **45**, and **32**).

Details of Individual Parts—Stepper Motor **22**

[0185] Stepper Motor **22** is shown in FIGS. **40** and **41**. It has a hexagonal Shaft **22-M1** that extends-through all the way from its rear surface to its front surface. The hexagonal shape of Shaft **22-M1** fits into hexagonal Hole **23-S1** of Spiral **23**, in manner such that the rotational position of Spiral **23** can be accurately controlled by Shaft **22-M1**, while allowing Spiral **23** to slide axially relative to Shaft **22-M1**.

[0186] Shaft **22-M1** has a centric Hole **22-M1-S1** that extends through the entire length of Shaft **22-M1**. Through Hole **22-M1-S1**, Spline **21** can be inserted in manner so that it can rotate relative to Hole **22-M1-S1** (see FIG. **32**).

[0187] Stepper Motor **22** has a Motor **22-M2**. In order to fix Motor **22-M2** to Cavity **20-S2** of Sleeve **20**, Motor **22-M2** has a Flange **22-M2-S1**, which has two Holes **22-M2-S2**. Each Hole **22-M2-S2** is for the insertion of a Bolt **32**, which is then threaded-on to a Threaded Hole **20-S3** of Sleeve **20** so as to secure Motor **22-M2** to Sleeve **20** (see FIGS. **34**, **35**, and **32**). In addition, Flange **22-M2-S1** also has two Cuts **22-M2-S3**, which allow Stepper Motor **22** to be inserted into Cavity **20-S2** without interfering with Spiral Positioning Pins **20-S10**.

[0188] If desired Stepper Motor **22** can be replaced with other means for controllably rotating a shaft, such as servo motor for example.

Details of Individual Parts—Spiral **23**

[0189] Spiral **23** is shown in FIGS. **42** to **47**; Drawings Set **1**: FIG. **42** shows a front-view of Spiral **23** and FIG. **43** shows a right-end-view of Spiral **23**; Drawings Set **2**: FIG. **45** shows a front-view of Spiral **23** (which is the left-view of Spiral **23** when Spiral **23** is oriented as shown in FIG. **46**), FIG. **47** shows a back-view of Spiral **23** (which is the right-view of Spiral **23** when Spiral **23** is oriented as shown in FIG. **46**), and FIGS. **44** and **46** show sectional-views of Spiral **23**. The hidden lines are not shown in FIGS. **44** to **47**.

[0190] Spiral **23** is shaped like cylinder that has a Front Outer Spiral Groove **23-S2** and Front Inner Spiral Groove **23-S4** shaped on its front-cylindrical-half (see FIG. **45**); and a Back Outer Spiral Groove **23-S6** and Back Inner Spiral Groove **23-S8** shaped on its back-cylindrical-half (see FIG. **47**).

[0191] During assembly, Follower Mechanism **24** (see FIGS. **51** to **53**) is threaded onto Spiral **23** by first inserting a Follower **24-S5** into each, Follower Insertion Slot **23-S3** and Follower Insertion Slot **23-S7** (see FIGS. **45** to **47**); and then rotating Follower Mechanism **24** so as to guide one Follower **24-S5** to its final assembled position in Front Outer Spiral Groove **23-S2**, and so as to

also guide the other Follower **24-S5** to its final assembled position in Back Outer Spiral Groove **23-S6**.

[0192] The profile of Front Outer Spiral Groove **23-S2** mirrors the profile of Back Outer Spiral Groove **23-S6** in a manner such that the rotation of Follower Mechanism **24** relative to Spiral **23** will cause “the axial movement applied to Follower Mechanism **24** due to the engagement of Front Outer Spiral Groove **23-S2** with its Follower **24-S5**” to be the same as “the axial movement applied to Follower Mechanism **24** due to the engagement of Back Outer Spiral Groove **23-S6** with its Follower **24-S5**”; so that Follower Mechanism **24** moves axially relative to Sleeve **20** in a manner so that Follower Disk **24-S1** is always parallel relative to the right-end surface of Sleeve **20**.

[0193] In order to be able rotate Spiral **23** and control the axial position of Spiral **23**, without having Front Outer Spiral Groove **23-S2** and Back Outer Spiral Groove **23-S6** apply any axial-forces on their Followers **24-S5**, when their Followers **24-S5** are at their neutral position; Spiral **23** also has a Front Inner Spiral Groove **23-S4** (see FIG. 45) and Back Inner Spiral Groove **23-S8** (see FIG. 47).

[0194] During assembly, Spiral **23** is threaded onto Sleeve **20** by first inserting a Spiral Positioning Pin **20-S10** (see FIGS. 34 and 35) into each, Pin Insertion Slot **23-S5** and Pin Insertion Slot **23-S9** (see FIGS. 44, 45, and 47); and then rotating Spiral **23** so as to guide one Spiral Positioning Pin **20-S10** to its final assembled position in Front Inner Spiral Groove **23-S4**, and so as to also guide the other Spiral Positioning Pin **20-S10** to its final assembled position in Back Inner Spiral Groove **23-S8**.

[0195] The engagement of the Spiral Positioning Pins **20-S10** with their Inner Spiral Grooves (Front Inner Spiral Groove **23-S4** or Back Inner Spiral Groove **23-S8** respectively), is used to control the axial position of Spiral **23** relative to Sleeve **20**.

[0196] Front Inner Spiral Groove **23-S4** is “parallel to” and has the same profile as Front Outer Spiral Groove **23-S2**; and Back Inner Spiral Groove **23-S8** is “parallel to” and has the same profile as Back Outer Spiral Groove **23-S6**; so that when Followers **24-S5** are at their neutral position, which is the position where they are horizontally aligned with Positioning Pins **20-S10**, the axial distance between “the Spiral Positioning Pins **20-S10** positioned in the Inner Spiral Grooves (Front Inner Spiral Groove **23-S4** and Back Inner Spiral Groove **23-S8**)” and “the Followers **24-S5** positioned in the Outer Spiral Grooves (Front Outer Spiral Groove **23-S2** and Back Outer Spiral Groove **23-S6**)” remains constant regardless of the rotational position of Spiral **23**; so that when Followers **24-S5** are at their neutral position, no axial force is applied to Followers **24-S5** due to the rotation of Spiral **23**.

[0197] Preferably, the “Spiral Grooves Profiles” of Spiral **23** (which are Front Outer Spiral Groove **23-S2**, Front Inner Spiral Groove **23-S4**, Back Outer Spiral Groove **23-S6**, and Back Inner Spiral Groove **23-S8**) have a lead angle that is small enough so as to prevent any axial forces applied on them by Followers **24-S5** and by Spiral Positioning Pins **20-S10** from rotating their Spiral **23**.

[0198] It is also recommended that Spiral **23** is at its most retracted position, which is a position where it cannot be rotated into its Sleeve **20** any further, when its CVT is shut-down. This is to prevent accidental/unintended rotation of Spiral **23** when its Stepper Motor **22** is turned-off, since Stepper Motor **22** has preferably some torque that prevents accidental/unintended rotation of Spiral **23** when it is turned-on.

[0199] If the Spirals **23** of a CVT are at their most retracted positions when their CVT are Shut-down, then the “Spiral Grooves Profiles” of the Cone(s) which is/are at its/their smallest Transmission Diameter when its/their CVT is/are Shut-down should differ from the “Spiral Grooves Profiles” of the Cone(s) which is/are at its/their largest Transmission Diameter when its/their CVT is/are Shut-down. The “Spiral Grooves Profiles” of a Spiral **23** can be obtained through experimentation or fundamental mathematics/engineering.

[0200] In order to rotate Spiral **23** with hexagonal-shaped Shaft **22-M1** of Stepper Motor **22**, Spiral **23** has a hexagonal Hole **23-S1** for Shaft **22-M1** (see FIGS. 42 and 43). The fit between Hole **23-S1**

and Shaft **22-M1** is such that Spiral **23** can smoothly slide axially relative Shaft **22-M1**, so as to allow Spiral **23** to move axially relative Sleeve **20**; while minimizing relative rotational movement between Spiral **23** and Shaft **22-M1**. And in order to limit how much Spiral **23** can retract into its Sleeve **20**, Spiral **23** has a Cover **23-S10** at its front-end (see FIG. **42**). The engagement of the rear surface of Cover **23-S10** with the front-end of Shaft **22-M1** will limit how much Spiral **23** can retract into its Sleeve **20**.

[0201] It is recommended that Spiral **23** is rotated when Torsion Plate **27** and Cone **25** are in their un-deflected rotational position relative to their Sleeve **20**. Torsion Plate **27** and Cone **25** are in their un-deflected rotational position relative their Sleeve **20** when no torque or an insignificant amount of torque is transmitted by Cone **25**. As such, it is recommended that Spiral **23** of a Cone **25** is rotated when its said Cone **25** is currently not used for torque transmission. Since when torque is transmitted by its Cone **25**, a significant torque is required in order to rotate Spiral **23**; because here Followers **24-S5** are rotationally deflected relative to Spiral Positioning Pins **20-S10** (Followers **24-S5** and Spiral Positioning Pins **20-S10** are not horizontally aligned), so that the “guided axial movement of Spiral **23** due to the engagement of the ‘rotating inner grooves (Front Inner Spiral Groove **23-S4** and Back Inner Spiral Groove **23-S8**)’ with Spiral Positioning Pins **20-S10**” will not match the “axial movement of Spiral **23** needed so that no axial forces are applied to its ‘rotationally deflected Followers **24-S5**’ by its ‘rotating outer grooves (Front Outer Spiral Groove **23-S2** and Back Outer Spiral Groove **23-S6**)’.”

[0202] It is also recommended that Spiral **23** is rotated to its “new ideal rotational position per the Transmission Diameter of its cone” immediately prior to the Transmission Diameter change of its cone, so the Spiral **23** is only not in its “ideal rotational position per the Transmission Diameter of its cone” immediately prior to the Transmission Diameter change of its cone.

Details of Individual Parts—Torsion Plate **27**

[0203] Torsion Plate **27** is shown in FIGS. **48** to **50**. FIG. **48** shows a front-view of Torsion Plate **27**, FIG. **49** shows a right-end-view of Torsion Plate **27**, and FIG. **50** shows a partial-front-view of Torsion Plate **27** when Torsion Plate **27** is oriented as shown in FIG. **49**.

[0204] Torsion Plate **27** has the shape of a “disk with a large hole” to which two rods and a spring connecting plate are fixed. Said disk is labeled as Disk **27-S1**, said large hole is labeled as Hole **27-S2**, said rods are labeled as Rods **27-S3**, and said spring connecting plate is labeled as Spring Connecting Plate **27-S4**.

[0205] The dimension of Hole **27-S2** is such that Torsion Plate **27** can smoothly and securely rotate on Sleeve **20** (see FIGS. **32** and **33**).

[0206] The dimension of Rods **27-S3** are such that they can be smoothly and securely inserted into Rod Holes **24-S3** of Follower Mechanism **24** (see FIGS. **51**, **52**, and **32**) and into Rod Holes **25-S4** of Cone **25** (see FIGS. **54**, **55**, and **32**). Rods **27-S3** also have sufficient length to stay engaged with Rod Holes **24-S3** and Rod Holes **25-S4** for all operating condition of their CVT. The engagement of Rods **27-S3** with Rod Holes **24-S3** and Rod Holes **25-S4** is used to constrain the rotational position of Follower Mechanism **24** and the rotational position of Cone **25** relative to Torsion Plate **27**, while allowing Follower Mechanism **24** and Cone **25** to move axially relative to Torsion Plate **27**.

[0207] Spring Connecting Plate **27-S4** has a Hole **27-S5** into which Mounting Rod **26-S2** of Spring **26** can be inserted and then secured using Locking Pin **31** (see FIGS. **48** to **50**, **37**, **39**, and **32**).

[0208] During assembly, the left-end of Spring **26** should be secured first by securing Mounting Rod **26-S1** to Spring Connecting Plate **20-S4** of Sleeve **20**. Then Rods **27-S3** of Torsion Plate **27** (which is already at its final assembled position on Sleeve **20**) should be inserted into Rod Holes **24-S3** of Follower Mechanism **24** (which is yet to be assembled to Sleeve **20**). Then while Follower Mechanism **24** is threaded onto Spiral **23** (which is also already at its final assembled position in Sleeve **20**), Spring Connecting Plate **27-S4** of Torsion Plate **27** will be rotated counter-clockwise towards Mounting Rod **26-S2** of Spring **26**; and when Follower Mechanism **24** has

reached its final assembled position, Spring Connecting Plate **27-S4** should be in a position where it can be fixed to Mounting Rod **26-S2** through the use of a Locking Pin **31** (see FIG. **32**), so as to fix the right-end of Spring **26** to Spring Connecting Plate **27-S4**.

Details of Individual Parts—Follower Mechanism **24**

[0209] Follower Mechanism **24** is shown in FIGS. **51** to **53**. FIG. **51** shows a front-view of Follower Mechanism **24**, FIG. **52** shows a left-end-view of Follower Mechanism **24**, and FIG. **53** shows a front-view of Follower Mechanism **24** when Follower Mechanism **24** is oriented as shown in FIG. **52**.

[0210] Follower Mechanism **24** comprises of a Follower Disk **24-S1**, which has a centric Spline Hole **24-S2** and two non-centric Rod Holes **24-S3**. And fixed to the rear surface of Follower Disk **24-S1** are two oppositely positioned Follower Attachment Bars **24-S4**; on which each a Follower **24-S5** is fixed. Each Follower **24-S5** is shaped like a short round bar.

[0211] Spline Hole **24-S2** is for the insertion of Spline **21**, on which Follower Disk **24-S1** can smoothly and securely rotate. Rod Holes **24-S3** are for the insertion of Rods **27-S3** of Torsion Plate **27**. One Follower Attachment Bar **24-S4** is used to position one Follower **24-S5** in Front Outer Spiral Groove **23-S2**; and the other Follower Attachment Bar **24-S4** is used to position the other Follower **24-S5** in Back Outer Spiral Groove **23-S6**.

[0212] The axial position of Follower Mechanism **24** depends on the rotational position of Follower Mechanism **24**, which in-turn depends on the amount of deflection of Spring **26** due to the torque that is transmitted by Cone **25**. The axial position of Follower Mechanism **24** depends on the rotational position of Follower Mechanism **24** because of the engagement of Follower **24-S5** with their respective Inner Spiral Grooves (Front Inner Spiral Groove **23-S4** and Back Inner Spiral Groove **23-S8**) of Spiral **23**.

[0213] Since Cone **25** can slide freely relative to Spline **21**, it is forced towards Follower Mechanism **24** by the tension of its Transmission Belt, so that Cone **25** is axially constrained relative to Follower Mechanism **24**. Because of this, the torque transmitted by Cone **25** will determine the axial position of Cone **25**, and as such the “Cone Diameter Increase Compensation” provided by Transmission Diameter Compensating Mechanism **19** of this section. If desired, Cone **25** can also be axially and rotatably constrained/fixed relative to Follower Mechanism **24** through gluing, screwed connections, etc.

Details of Individual Parts—Cone **25**

[0214] Cone **25** is shown in FIGS. **54** to **56**. It has a Front Cover **25-S1**, which has a centric Spline Hole **25-S2**. Spline Hole **25-S2** is for the insertion of Spline **21**, on which Front Cover **25-S1** can smoothly and securely rotate and slide axially.

[0215] Cone **25** also has a Rear Cover **25-S3**. Rear Cover **25-S3** has a centric Spline Hole **25-S5**. Spline Hole **25-S5** is for the insertion of Spline **21**, on which Rear Cover **25-S3** can also smoothly and securely rotate and slide axially. In addition, Rear Cover **25-S3** also has two non-centric Rod Holes **25-S4**, which are for the insertion of Rods **27-S3** of Torsion Plate **27** as described previously. Method of Changing the Transmission Diameters of the Cones in Increments Specifically Selected for Each Transmission Diameter and Transmission Ratio

[0216] For the Driving Cone and Driven Cone, there are variation in “Transmission Belt Teeth Compression” at different Transmission Diameters; since for a Smaller Transmission Diameter the pressure of the Transmission Belt is dispersed over a smaller area than for a Larger Transmission Diameter.

[0217] Said “Transmission Belt Teeth Compression” variation can affect the Transmission Diameter of a Transmission Belt. A larger compression will compress the Teeth of a Transmission Belt more, so that its Transmission Belt is positioned at a lower height, so as to reduce its Transmission Diameter. Here changing the Transmission Diameter of a Cone at fixed interval, as is ideal when there is no “Transmission Belt Teeth Compression” variation at different Transmission Diameters, can cause tooth misalignment during re-engagement of a Torque Transmitting Member

with its Transmission Belt.

[0218] In addition, for a CVT for which the Tensioning Pulley is positioned on the slack-side of its Transmission Belt, such as the CVT shown in FIG. 5; for the Driven Cone, there are also variation in “slack-side Transmission Belt stretching” at different Transmission Ratios (which depends on the “Transmission Diameter of the Driving Cone” and the “Transmission Diameter of the Driven Cone”). This is especially true if the axial positions of the Cones are changed independently as according to the method described in European Patent EP2764279; since for this method, slack needs to be provided or removed depending on whether the axial position of the Cones are “aligned” or “not aligned” for their current Transmission Ratio. And this can cause the tension in the slack side of the Transmission Belt to vary significantly between different Transmission Ratios; since the angles at which the Transmission Belt approaches and leaves its Tensioning Pulley can vary significantly between when slack needs to be provided and when slack needs to be removed.

[0219] In order to eliminate/minimize “tooth misalignment during re-engagement” due to “Transmission Belt Teeth Compression” variation at different Transmission Diameters, the following method of can be used for the Driving Cone: Changing/selecting the Transmission Diameters of the Driving Cone in increments that depends on the current Transmission Diameter of the Driving Cone. For example, the Transmission Diameters of a Driving Cone can be selected as follows: 4.00 inches, 4.25 inches (0.25 in. increment increase), 4.4375 inches (0.1875 in. increment increase), 4.5625 inches (0.125 in. increment increase), and so-forth. The ideal profile for changing/selecting the Transmission Diameters of the Driving Cone, which depends on the axial position of said Driving Cone, can be obtained experimentally or mathematically.

[0220] For the Driven Cone, the ideal “axial position changing increment”/“Transmission Diameter changing increment” of the Driven Cone depends on both the “Transmission Diameter of the Driving Cone” and the “Transmission Diameter of the Driven Cone”. Since both the “Transmission Diameter of the Driving Cone” and the “Transmission Diameter of the Driven Cone” determine the tension in the slack-side of the Transmission Belt; since they both determine whether the Cones are “aligned” or “not aligned”, and the angles at which the Transmission Belt approaches and leaves its Tensioning Pulley.

[0221] In order to eliminate/minimize the “tooth misalignment during re-engagement” due to “Transmission Belt Teeth Compression” variation and “slack-side Transmission Belt stretching” variation at different Transmission Diameters, the following method can be used for the Driven Cone: Changing/selecting the Transmission Diameters of the Driven Cone in increments that depends on the “current Transmission Diameter of the Driving Cone” and the “current Transmission Diameter of the Driven Cone”.

[0222] An example of how the Transmission Diameter of a Driven Cone can be changed so as to minimize “tooth misalignment during re-engagement”, is as follows: a) when the Transmission Diameter of the Driving Cone is 4.00 inches and the Transmission Diameter of the Driven Cone is also 4.00 inches, then the Transmission Diameter of the Driven Cone is increased by a 0.25 inch increment to 4.25 inches; b) when the Transmission Diameter of the Driving Cone is 4.25 inches and the Transmission Diameter of the Driven Cone is also 4.25 inches, the Transmission Diameter of the Driven Cone is increased by a 0.1875 inch increment to 4.4375 inches; and c) when the Transmission Diameter of the Driving Cone is 4.4375 inches and the Transmission Diameter of the Driven Cone is also 4.4375 inches, the Transmission Diameter of the Driven Cone is increased by a 0.125 inch increment to 4.5625 inches; and so-forth. The ideal profile for changing/selecting the Transmission Diameters of the Driven Cone, which depends on the axial position of the Driving Cone and the axial position of said Driven Cone, can be obtained experimentally or mathematically.

[0223] It is possible to change the Transmission Ratio in a manner so that the “Transmission Diameter of the Driving Cone” is always the same for a given “Transmission Diameter change” of the driven Cone. If this is the case, then the “Transmission Diameter of the Driven Cone” should

also be changed/selected in increments that only depend on the “Transmission Diameter of the Driven Cone”.

[0224] The method of this section requires that that the “fixed increment movements” of an “axial position changing mechanism” are converted into “variable increment movements”. A mechanism that can be used to convert the “fixed increment movements” of an “axial position changing mechanism” into “variable increment movements” is described in the section below.

Mover Sliding Plate Mechanism for Converting Fixed Interval Movements to Required Interval Movement for Moving a Cone

[0225] The axial position of a cone can be changed quickly and accurately using a “Transmission Ratio Changing Mechanism” described in the “Transmission Ratio Changing Mechanisms” section of U.S. Pat. No. 9,958,064, preferably a “Lever Indexing Mechanism 2”.

[0226] The movements provided by a “Transmission Ratio Changing Mechanism” is fixed, while the required axial movements for a Cone from one Transmission Diameter that allows for optimal engagement to the next Transmission Diameter that allows for optimal engagement might change with the change in the Transmission Diameter of its Cone. If so, in order to have a “Transmission Ratio Changing Mechanism” provide the required amount of axial movements for a Cone, a “Mover Sliding Plate Mechanism”, described in this section can be used.

[0227] A “Mover Sliding Plate Mechanism” that can be used to control the axial position of a Cone is shown as a partial side-view in FIGS. 57 and 58, as a partial front-view in FIG. 59, and as a partial top-view in FIG. 60. It comprises of two parallel Sliding Plates 33 that are constrained so that they can only be moved in the up-&-down directions shown in FIG. 57. The position of one Sliding Plate 33 relative to the other is fixed through the use of a Connector Plate 34 so that the Sliding Plates 33 are always aligned such that the Mover Pins 35, which end portions slide in Slot 33-S1 of a Sliding Plate 33, are always perpendicular to Sliding Plates 33.

[0228] In FIGS. 57 to 60, Connector Plate 34 is welded to Sliding Plates 33 for simplicity in describing the mechanism. Obviously here and in all other parts of this description where applicable, other methods for connecting/joining/attaching can be used. For example, here fasteners such as bolts, nuts, locking rings, etc., can be used for ease of assembly and disassembly and to prevent warping; or gluing can also be used.

[0229] A Mover Rod 36 is shown in FIGS. 61 to 63. Mover Rod 36 has a hole near its rear-end, into which a Mover Pin 35 is tightly inserted; in a manner so that Mover Pin 35 extends to the sides of Mover Rod 36, and so that Mover Pin 35 is fixed relative Mover Rod 36. And at the front-end of Mover Rod 36, a Mover Connector 37 is fixed.

[0230] Mover Pin 35 is a long slender cylinder that has a Locking Ring Groove shaped near each of its ends (see FIG. 62).

[0231] Mover Connector 37 is a slender bar that has a hole through which the front-end of Cone 38 (which is sliding on a Spline 44) can be slid in and secured for axial (but not rotational) movements relative to Mover Connector 37 using a Locking Ring 39. Because of Mover Rod 36, the axial position of Cone 38 is dependent on the axial position of Mover Pin 35.

[0232] The engagement of Mover Pin 35 with Slots 33-S1 of Sliding Plates 33 are used to convert the “Fixed Interval Up-&-Down Movements” of Sliding Plates 33 into “Variable Interval Axial Movement” of Cone 38.

[0233] FIGS. 57 and 58 might not accurately show the shape of Slots 33-S1. The exact shape for Slots 33-S1 can be obtained through experimentation, mathematics, engineering, and so forth.

[0234] In order to limit the movements of the Mover Pin 35, and hence also the movements of the Mover Rod 36, to horizontal movements (see FIG. 57), two parallel Horizontal Movement Plates 40 that each has a Horizontal Slot are used (see FIGS. 58 to 60). Horizontal Movement Plates 40 are aligned and fixed to a non-moving part of the CVT, such as the housing of the CVT for example, in manner such that Mover Pins 35, which end portions each slide in a Horizontal Slot of a Horizontal Movement Plate 40, are always horizontal.

[0235] Mover Rod **36** is positioned between the parallel Sliding Plates **33** and the parallel Horizontal Movement Plates **40** (see FIGS. **59** and **60**). In order to secure Mover Rod **36** to Sliding Plates **33** and Horizontal Movement Plates **40**, two Locking Rings **41** that sandwich the Horizontal Movement Plates **40** are used (see FIGS. **59** and **60**); here each Locking Ring **41** is inserted into a Locking Ring Groove of Mover Pin **35** (see FIG. **62**).

[0236] It is recommended that friction between the slots of “Sliding Plates **33** and Horizontal Movement Plates **40**” and “Mover Pins **35**” is minimized; and it is also recommended that friction between “Locking Rings **41**” and “Sliding Plates **33**” is minimized; this can be achieved by making or coating Sliding Plates **33** and Horizontal Movement Plates **40** with a low friction material, or submerging the mechanism in oil.

[0237] In order to move the Sliding Plates **33** in the up & down directions shown in FIG. **57**, a “gear rack drive” can be used. In FIGS. **57** & **60**, a “gear rack drive” that has a Gear **42** and a Gear Rack **43**, which is fixed to a Sliding Plate **33**, is used to move the Sliding Plates **33** in the up & down directions shown in FIG. **57**. A transmission ratio changing mechanism described in the “Transmission Ratio Changing Mechanisms” section of U.S. Pat. No. 9,958,064, preferably a “Lever Indexing Mechanism 2”, can be used to provide the rotational input for the Gear **42**. The rotational input for the Gear **42** can also be provided by other means such as a stepper motor for example.

[0238] Each Sliding Plate **33** has two parallel Vertical Guides **45** (see FIGS. **58** & **60**), which sandwich the side surfaces of Sliding Plates **33**. Vertical Guides **45** are used to constrain the movements of Sliding Plates **33** to the up & down directions. It is preferably to keep friction between the side surfaces of the Sliding Plates **33** and the constraining surfaces of Vertical Guides **45** to a minimum.

Transmission Belt Constraining and Tensioning Method

[0239] The “Transmission Belt Constraining Method” for a CVT that comprises of a “Cone with One Torque Transmitting Member” that is coupled by a Transmission Belt to another “Cone with One Torque Transmitting Member” is shown in FIGS. **64** and **65**. If desired, each “Cone with One Torque Transmitting Member” can be replaced with a “Cone with One Torque Transmitting Member and One Non-Torque Transmitting Member”.

[0240] The main purposes/advantages of the “Transmission Belt Constraining Method” are as follows: a) the Constrainer Wheels **48-M1**, which are on the slack-side of Transmission Belt **50**, will partially or fully “remove and/or provide slack” due changing the axial position of the Cones independently as per the method described in European Patent EP2764279; and b) the Constrainer Wheel **48-M1** on the tense-side of Transmission Belt **50** will prevent teeth ratcheting at the “exit point”, which is the point where engagement between the teeth of Torque Transmitting Member **46-M1** and the teeth of Transmission Belt **50** ends; teeth ratcheting will most likely occur at the point of exit; this can allow for the usage of less pre-tension in the slack side of the Transmission Belt **50**.

[0241] The CVT shown in FIGS. **64** and **65** comprise of a Driving Cone **46** (which has a Torque Transmitting Member **46-M1**) that is coupled by a Transmission Belt **50** to a Driven Cone **47** (which has a Torque Transmitting Member **47-M1**); in a manner so that the apex of Driving Cone **46** and the apex of Driven Cone **47** point in opposite directions (see configuration shown in FIG. **2**). In order to constrain the slack-side of Transmission Belt **50**, a Constrainer **48** is used.

Constrainer **48** has two Constrainer Wheels **48-M1**; one for pushing Transmission Belt **50** against Driving Cone **46**, and one for pushing Transmission Belt **50** against Driven Cone **47**. The Constrainer Wheels **48-M1** should push Transmission Belt **50** against their Cones hard enough, so that the “the portions of Transmission Belt **50**” and “the portions of its Torque Transmitting Member or Non-Torque Transmitting Member” at the point of contact with Constrainer Wheels **48-M1** are always fully pushed against their Cone regardless how large the centrifugal forces are.

[0242] Constrainer **48** should be mounted relative to its CVT so that it is pushed towards its CVT as described in the paragraph above. Here one or several spring(s) can be used. Constrainer **48**

should also be allowed to move sideways, so that it can slide left or right as required during Transmission Ratio Change. But, Constrainer **48** should not be allowed to tilt, so as to ensure that Constrainer **48** is always positioned in the same position for each Transmission Ratio. Constrainer **48** should also be mounted so that its axial position relative to Transmission Belt **50** is fixed. [0243] In order to provide the pretension for Transmission Belt **50**, a Tensioning Pulley **51** is used. Tensioning Pulley **51** is mounted on Constrainer **48** so that it can only slide up and down relative to Constrainer **48**; and Tensioning Pulley **51** is pushed up relative to Constrainer **48** by a spring or other means, so that it applies tension to Transmission Belt **50**.

[0244] In order to constrain the tense-side of Transmission Belt **50**, a Constrainer Wheel **49** is used. Constrainer Wheel **49** pushes Transmission Belt **50** against Driving Cone **46** and Driven Cone **47**. Constrainer Wheel **49** should push Transmission Belt **50** against their Cones hard enough, so that the “the portions of Transmission Belt **50**” and “the portions of its Torque Transmitting Member or Non-Torque Transmitting Member” at the point of contact with Constrainer Wheels **48-M1** are always fully pushed against their Cone regardless how large the centrifugal forces are.

[0245] Constrainer Wheel **49** should be mounted relative to its CVT so that it is pushed towards its CVT as described in the paragraph above. Here one or several springs can be used. Constrainer Wheel **49** should also be allowed to move sideways, so that it can slide left or right as required during Transmission Ratio Change. Constrainer Wheel **49** should also be mounted so that its axial position relative to Transmission Belt **50** is fixed.

“Tensioning and Constraining Mechanism” for “Transmission Belt Constraining and Tensioning Method”

[0246] Described In this section is a “Tensioning and Constraining Mechanism” that can be used for the “Transmission Belt Constraining and Tensioning Method” of the previous section.

[0247] The “Tensioning and Constraining Mechanism”, which is labeled as “Tensioning and Constraining Mechanism **52**”, is shown in FIGS. **66** and **67**. It has a Constrainer **53**, which performs the function of Constrainer **48** of FIGS. **64** and **65**. Constrainer **53**, which is shown in detail in FIGS. **68** to **71**, has two Holes **53-S2** that are used to mount two Constrainer Wheels **53-M1**, through the use of a Shafts **53-M2** and Locking Rings **53-M3**. Locking Rings **53-M3** as well as other fasteners of this disclosure can be replaced with other fasteners, such as replacing nuts-and-bolts with locking rings and vice-versa for example. Constrainer Wheels **53-M1** perform the functions of Constrainer Wheels **48-M1** of FIGS. **64** and **65**. Each Constrainer Wheel **53-M1** has a bearing which extends beyond the side surfaces of its pulley and engage with the sidewalls that cradle it (see FIG. **67**).

[0248] Constrainer **53** also has two Vertical Slots **53-S3** that are used to slideably insert a Tensioning Pulley Slider Mechanism **54**; in a manner such that Tensioning Pulley Slider Mechanism **54** is only allowed to slide vertically relative to Constrainer **53**. Tensioning Pulley Slider Mechanism **54** has a Tensioning Pulley **54-M2**, which performs the function of Tensioning Pulley **51** of FIGS. **64** and **65**. Tensioning Pulley **54-M2** has a bearing which extends beyond the side surfaces of its pulley and engage with the sidewalls that cradle Tensioning Pulley **54-M2** (see FIGS. **67** and **73**).

[0249] In order to push Constrainer **53** towards its Cones, Springs **55** are used. Springs **55** provide the compression force that pushes Constrainer **53** towards Constrainer Wheel Mounting Mechanism **59**, so as to force Constrainer **53** and Constrainer Wheel Mounting Mechanism **59** toward the surfaces of their Cones. In order to achieve this, a Constrainer Push Plate **57**, which is shown in detail in FIGS. **75** to **77**, is positioned at the bottom-end of Springs **55** and is fixed vertically relative to Springs **55**; so that Springs **55** are positioned between Constrainer **53** and Constrainer Push Plate **57**, so as to force them apart, and as a result push Constrainer Push Plate **57** towards Constrainer Wheel Mounting Mechanism **59**.

[0250] Constrainer **53** (see FIG. **71**), Constrainer Push Plate **57** (see FIG. **76**), and Constrainer Wheel Mounting Mechanism **59** (see FIG. **81**), each has “two sleeves”, or “two holes”, or “two

sleeves and two holes” through which a Connecting Rod **58** is inserted. Connecting Rod **58** is shown in FIG. **78**, it is slender round rod that has a Connecting Rod Bottom **58-S1** and a Threaded End **58-S2**.

[0251] The purpose of Connecting Rod **58** (which is fixed to a Base Plate **61**) is to: a) fix the “bottom-end vertical position” and “horizontal position” of Constrainer Push Plate **57** relative to Base Plate **61**; b) fix the “top-end vertical position” and “horizontal position” of Constrainer Wheel Mounting Mechanism **59** relative to Base Plate **61**; and c) fix the horizontal position of Constrainer **53** relative to Base Plate **61**, while allowing Constrainer **53** to slide vertically relative Constrainer Push Plate **57** and Constrainer Wheel Mounting Mechanism **59**, as forced by Springs **55**.

[0252] The “bottom-end vertical position” of Constrainer Push Plate **57** relative to Connecting Rod **58** is fixed through the engagement of Connecting Rod Bottoms **58-S1** with the bottom-surface of Constrainer Push Plate **57** (see FIGS. **66** and **67**); and the “top-end vertical position” of Constrainer Wheel Mounting Mechanism **59** relative to Connecting Rod **58** is fixed through the engagements of a Nuts **60** (which are threaded onto Threaded Ends **58-S2**) and the top surface of Constrainer Wheel Mounting Mechanism **59**.

[0253] The engagement of Connecting Rod **58** with the holes and/or sleeves of Constrainer **53**, Constrainer Push Plate **57**, and Constrainer Wheel Mounting Mechanism **59** will fix the horizontal position of Constrainer **53**, Constrainer Push Plate **57**, and Constrainer Wheel Mounting Mechanism **59** relative to Connecting Rod **58**, and as such also relative to Base Plate **61**.

[0254] In order to allow Constrainer **53** to move sideways as required during Transmission Ratio Change, while preventing Constrainer **53** from tilting as required by the “Transmission Belt Constraining and Tensioning Method”, the bottom-ends of Connecting Rod Bottoms **58-S1** are each fixed to a Base Plate **61**, which is constrained to slide in the horizontal direction relative to a frame by Constraining Surfaces **62**. In FIGS. **66** and **67**, the bottom-ends of Connecting Rod Bottoms **58-S1** are each fixed to Base Plate **61** through gluing; obviously other methods of fixing can be used, such as by using fasteners such as bolts, nuts, threaded holes, etc.

[0255] In order to allow Constrainer Wheel **59-M2** of Constrainer Wheel Mounting Mechanism **59** to move sideways relative to Constrainer **53**, so that it can slide left or right as required during Transmission Ratio Change, Slider Blocks **59-M4** are attached to the ends of a Shaft **59-M3** of Constrainer Wheel **59-M2**. Slider Blocks **59-M4** are then inserted into Slots **59-S3** (see FIGS. **79** to **81**) that allow Constrainer Wheel **59-M2** to slide sideways.

Details of Individual Parts—Constrainer **53**

[0256] Constrainer **53** is shown in FIGS. **68** to **71**. It has two Side Plates **53-S1** that are joined by two Bottom Plates **53-S4**, which are separated by Vertical Slots **53-S3** (see FIG. **70**). Each Side Plate **53-S1** has two Holes **53-S2** and a Vertical Slot **53-S3**. Each pair of Holes **53-S2** from two facing Side Plates **53-S1** are used for inserting a Shaft **53-M2** on which a Constrainer Wheel **53-M1** is mounted (see FIGS. **66** and **67**). And Vertical Slots **53-S3** are used to slideably insert Tensioning Pulley Slider Mechanism **54** as shown in FIGS. **66** and **67**.

[0257] In addition, each Side Plate **53-S1** also has a Sleeve **53-S6**, which is used to insert a Connecting Rod **58**, as shown in FIGS. **66** and **67**.

[0258] Each Bottom Plate **53-S4** has a Spring Holder **53-S5** shaped on its bottom-surface. Spring Holders **53-S5** are used to maintain the position of the top-end of Springs **55**.

Details of Individual Parts—Tensioning Pulley Slider Mechanism **54**

[0259] Tensioning Pulley Slider Mechanism **54** is shown in FIGS. **72** to **74**. It has a Frame **54-M1** on which “a Shaft **54-M3** with a Tensioning Pulley **54-M2**” is mounted.

[0260] Frame **54-M1** has two Side Plates that are joined by a Bottom Plate (see FIG. **73**). Each said Side Plate has Hole **54-M1-S1** for Shaft **54-M3**. And said Bottom Plate has Spring Holder **54-M1-S2** shaped at the center of its bottom-surface. Said Spring Holder **54-M1-S2** is used to maintain the position of the top-end of Spring **56**.

[0261] Tensioning Pulley **54-M2** has a bearing which extends beyond the side surfaces of its pulley

and engage with the sidewalls that cradle Tensioning Pulley **54-M2**. And the axial position of Shaft **54-M3** is held in place by side walls of Vertical Slot **53-S3**.

Details of Individual Parts—Constrainer Push Plate **57**

[0262] Constrainer Push Plate **57** is shown in FIGS. **75** to **77**. On the top-surface of Constrainer Push Plate **57** two Spring Holders **57-S1** and a Spring Holder **57-S2** are shaped. Spring Holders **57-S1** are used to maintain the position of the bottom-ends of Springs **55**; and Spring Holder **57-S2** is used to maintain the position of the bottom-end of Spring **56**.

[0263] In addition, Constrainer Push Plate **57** also has two Holes **57-S3**, which each will be used to insert a Connecting Rod **58**, as shown in FIGS. **66** and **67**.

Details of Individual Parts—Constrainer Wheel Mounting Mechanism **59**

[0264] Frame **59-M1** of Constrainer Wheel Mounting Mechanism **59** is shown in FIGS. **79** to **81**. Frame **59-M1** has a Top Plate **59-S1** on which two parallel Side Plates **59-S2** are fixed. Each Side Plate **59-S2** has a Slot **59-S3**; and to each Side Plate **59-S2**, a Sleeve **59-S4** is fixed. Each Sleeve **59-S4** has a hole that is concentric with a Hole **59-S5** of Top Plate **59-S1**.

[0265] Constrainer Wheel **59-M2** (which has a bearing which extends beyond the side surfaces of its pulley and engage with Slider Blocks **59-M4**), its Shaft **59-M3**, and its Slider Blocks **59-M4** are shown in FIGS. **82** and **83**. In the assembled state of Constrainer Wheel Mounting Mechanism **59** each Slider Block **59-M4** is positioned inside a Slot **59-S3** (see FIGS. **79** to **81**).

Alternate “Tensioning and Constraining Mechanisms”

Tensioning and Constraining Mechanism with No Tensioning Pulley

[0266] A “Tensioning and Constraining Mechanism” that can be used instead of “Tensioning and Constraining Mechanism **52**” is a “Tensioning and Constraining Mechanism **52A**”, which is shown in FIGS. **84** and **85**.

[0267] “Tensioning and Constraining Mechanism **52A**” is identical to “Tensioning and Constraining Mechanism **52**”, except that it does not have a Tensioning Pulley Slider Mechanism **54**, and as such no Tensioning Pulley **54-M2**.

[0268] “Tensioning and Constraining Mechanism **52A**” can work if the “variation in slack in the slack-side of the Transmission Belt” due to “variation in pulling load” and/or “slack being removed/provided at different Transmission Ratios” can be kept acceptable with sufficient pre-tension and/or through the use of a Transmission Diameter Compensating Mechanism, such as Transmission Diameter Compensating Mechanism **10** or Transmission Diameter Compensating Mechanism **10A**.

“Tensioning and Constraining Mechanism” for CVT 6 of U.S. Pat. No. 9,651,123

[0269] A CVT 6 uses two CVT 4's for which the Driving Cones of said CVT 4's are mounted on a common spline/shaft and the Driven Cones of said CVT 4's are also mounted on a common spline/shaft. For a CVT 6, the tension in the Transmission Belt for the CVT 4 for which the axial position of a Cone is changed can be reduced. In order to allow for this, both the Support Pulley and Tensioning Pulley of each CVT 4 need to provide/remove slack as needed during axial position changing of a Cone (refer to U.S. Pat. No. 9,651,123).

[0270] A CVT 4 to which the “Transmission Belt Constraining and Tensioning Method” is applied is shown in FIG. **86**. It is identical to the “Transmission Belt Constraining and Tensioning Method” shown in FIG. **65**, except for the following: a) It uses Tensioning Pulley **51A**, which is configured so that it can remove and provide more slack than Tensioning Pulley **51**; b) It uses a Transmission Belt **50A**, which is longer than Transmission Belt **50** so that more slack can be removed and provided; and c) It uses a Support Pulley **49A** instead of a Constrainer Wheel **49**.

[0271] Support Pulley **49A** is mounted so that it can move sideways as required during Transmission Ratio changes, and it is forced towards the surface of its Cones by a spring, gravity, or other means. Also unlike a Constrainer Wheel **49A**, when its CVT 4 is used to transmit torque, Support Pulley **49A** should not be pushing “sections of its Transmission Belt that are resting on the surfaces of their Cones” towards the “surfaces of said Cones”. Here, Support Pulley **49A** should be

forced away from the surfaces of its Cone by the tension in its Transmission Belt until its Mounting **63** hits Stops **64**; in a configuration so that Support Pulley **49A** has sufficient range to move “towards” and “away from” the surfaces of its Cones when the tension in its Transmission Belt is reduced (refer to U.S. Pat. No. 9,651,123).

[0272] A “Tensioning and Constraining Mechanism” for the “Transmission Belt Constraining and Tensioning Method” for a CVT 4, as shown in FIG. **86**, is a “Tensioning and Constraining Mechanism **52B**”, which is shown in FIGS. **87** and **88**.

[0273] “Tensioning and Constraining Mechanism **52B**” is identical to “Tensioning and Constraining Mechanism **52**”, except that: a) Constrainer Wheel Mounting Mechanism **59** is replaced with Support Wheel Mounting Mechanism **59A**; b) it uses two Connecting Rods **58A** instead of two Connecting Rods **58**; c) it has an additional Top Plate **65** which “vertical position” and “horizontal position” is fixed relative Connecting Rod **58A**; and d) it has two Springs **66**, which force Support Wheel Mounting Mechanism **59A** away from Top Plate **65**.

[0274] Support Wheel Mounting Mechanism **59A** is identical to Constrainer Wheel Mounting Mechanism **59**, except that two Spring Holders **59A-S1** are shaped on its top-surface (see FIGS. **87** and **88**). Spring Holders **59-S1** are used to maintain the position of the bottom-end of Spring **66**. Each Spring Holder **59-S1** is concentric with the hole for a Connecting Rod **58A**.

[0275] A Connecting Rod **58A** is shown in FIG. **89**. Connecting Rods **58A** are identical to Connecting Rods **58**, except that: a) they are longer in order to account for Springs **66** and the slack that needs to be removed when the tension in their Transmission Belt is reduced; b) they have a Top-End **58A-S1**, which has a smaller diameter than their main shaft (which are positioned below them); c) their Top-End each have a Locking Ring Groove **58A-S2** for a Locking Ring **67**.

[0276] A Top Plate **65** is shown in FIGS. **90** to **92**. Top Plate **65** has a Plate **65-S1** which has two Holes **65-S2**. Concentric with Holes **65-S2** are two Spring Holders **65-S3**; each Hole **65-S2** extends through the entire depth of Plate **65-S1** and its Spring Holder **65-S3**.

[0277] Each Hole **65-S2** is for the insertion of a Top-End **58A-S1** of a Connecting Rod **58A**; it should have the same dimension as Top-End **58A-S1** plus some required tolerances. Since the diameters of the main shafts of Connecting Rods **58A** are larger than the diameters of Top-Ends **58A-S1**, and such also the diameters of Holes **65-S2**; the top-surfaces of the main shafts of Connecting Rods **58A** are resting on the bottom surfaces of Spring Holders **65-S3** (see FIGS. **87** and **88** for assembly, and FIGS. **90** and **91** for labeling of individual shapes of Top Plate **65**). Here the engagement of the bottom-surfaces of Spring Holders **65-S3** with the main shafts of Connecting Rods **58A** will prevent Top Plate **65** from moving downwards relative to Connecting Rods **58A**. And the engagement of the top-surface of Plate **65-S1** with Locking Ring **67**, will prevent Top Plate **65** from moving upwards relative to Connecting Rods **58A** (see FIGS. **87** and **88** for assembly, and FIGS. **90** and **91** for labeling of individual shapes of Top Plate **65**).

[0278] Top Plate **65** also has a Stop **65-S4**, which is shaped on the bottom-surface of Plate **65-S1**. Stop **65-S4** performs the same function as Stops **64** of FIG. **86**.

PREFERRED EMBODIMENT OF THE INVENTION (BEST MODE)

[0279] The preferred CVT of this disclosure is a CVT 6 (refer to U.S. Pat. No. 9,651,123) for which each CVT 4 uses the configuration and items shown in FIG. **86**. For a CVT 6, the tension in the transmission belt for the CVT 4 for which the axial position of a cone is changed is reduced

[0280] For the configuration shown in FIG. **86**, the tensioning pulley, which is labeled as Tensioning Pulley **51A**, is positioned on the slack side of the transmission belt; this will maintain the tension in the slack-side of Transmission Belt **50A** almost constant in order to ensure smooth re-engagement of Driven Cone **47**.

[0281] And in order to compensate for an increase in “Transmission Belt Teeth Compression” and an increase in “stretching of the tense side of the Transmission Belt” due to an increase in tension in the tense side of the Transmission Belt when an increased amount of torque is being transmitted by the CVT, so as to ensure smooth re-engagement of Driving Cone **46**; Driving Cone **46** uses

Transmission Diameter Compensating Mechanism **19** described in the “Transmission Diameter Compensating Mechanism with Adjustable Spiral” section of this disclosure.

[0282] In addition, the configuration shown in FIG. **86** also uses Constrainer **48** on the slack of its transmission belt in order to reduce/minimize changes of the “angles at which the Transmission Belt approaches and leaves its Tensioning Pulley” due to whether the Cones are “aligned and “not aligned” for their current Transmission Ratio, and/or due to other operational variations of the CVT; which will reduce variation in tension in the slack side of the Transmission Belt. In addition, Constrainer **48** also prevents teeth ratcheting at the point where engagement between the pulling Torque Transmitting Member and its Transmission Belt ends.

[0283] And in order to allow for the tension in the Transmission Belt of the CVT **4** for which the axial position of a Cone is changed to be reduced, the configuration shown in FIG. **86** also uses a Support Pulley **49A** which can provide/remove “transmission belt slack” as needed during axial position changing of a Cone of its CVT **4** (refer to U.S. Pat. No. 9,651,123).

[0284] The preferred CVT of this disclosure also uses the “Method of changing the Transmission Diameters of the Cones in increments specifically selected for each Transmission Diameter and Transmission Ratio” in order to select the axial position changing increments/distances of its Cones. The “Method of changing the Transmission Diameters of the Cones in increments specifically selected for each Transmission Diameter and Transmission Ratio” is the preferred method of selecting the axial position changing increments/distances of the Cones of the CVT's of this disclosure.

ALTERNATE EMBODIMENTS

[0285] For all CVT's described in this disclosure a “Cone with one Torque Transmitting Member” can be replaced with a “Cone with two oppositely positioned Torque Transmitting Members”. FIG. **93** shows a CVT that is identical to the CVT shown in FIG. **5**; except that Driving Cone **5** is replaced with a Driving Cone **5A**, and Driven Cone **6** is replaced with a Driving Cone **6A**.

[0286] Driving Cone **5** and Driven Cone **6** are each a “Cone with one Torque Transmitting Member”; while Driving Cone **5A** and Driven Cone **6A** are each a “Cone with two oppositely positioned Torque Transmitting Members”. The Torque Transmitting Members of Driving Cone **5A**, which are substantially, but not necessarily exactly opposite, are labeled as Torque Transmitting Members **5A-M1**); and the Torque Transmitting Members of Driven Cone **6A**, which are also substantially, but not necessarily exactly opposite, are labeled as Torque Transmitting Members **6A-M1**).

[0287] FIG. **94** shows a CVT that is identical to the CVT shown in FIG. **64**; except that Driving Cone **46** is replaced with a Driving Cone **46A**, and Driven Cone **47** is replaced with a Driving Cone **47A**.

[0288] Driving Cone **46** and Driven Cone **47** are each a “Cone with one Torque Transmitting Member”; while Driving Cone **46A** and Driven Cone **47A** are each a “Cone with two oppositely positioned Torque Transmitting Members”. The Torque Transmitting Members of Driving Cone **46A**, which are substantially but not necessarily exactly opposite, are labeled as Torque Transmitting Members **46A-M1**); and the Torque Transmitting Members of Driven Cone **47A**, which are also substantially but not necessarily exactly opposite, are labeled as Torque Transmitting Members **47A-M1**).

CONCLUSION, RAMIFICATIONS, AND SCOPE

[0289] While my above description contains many specificities, these should not be construed as limitations on the scope, but rather as an exemplification of one or several embodiment(s) thereof. Many other variations are possible.

[0290] Accordingly, the scope should be determined not by the embodiment(s) illustrated, but by the appended claims and their legal equivalents.

Claims

1. A toothed CVT using a Driving Cone, which is a Cone with One Torque Transmitting Member, that is coupled by a Transmission Belt to a Driven Cone, which is also a Cone with One Torque Transmitting Member, for which the re-engagement of the teeth of said Torque Transmitting Members with the teeth of said Transmission Belt is improved by compensating for an increase in Transmission Belt Teeth Compression and an increase in stretching of the tense side of said Transmission Belt due to an increase in tension in the tense side of said Transmission Belt, when an increased amount of torque is being transmitted, comprising: a) said Driving Cone; b) said Driven Cone; c) said Transmission Belt; d) a Tensioning Pulley, which is positioned on the slack side of said Transmission Belt, so as to maintain the tension in the slack-side of said Transmission Belt almost constant; e) a Transmission Diameter Compensating Mechanism, which increases the Transmission Diameter of said Driving Cone when the torque being pulled by said Driving Cone is increased, in order to compensate for an increase in Transmission Belt Teeth Compression and an increase in stretching of the tense side said Transmission Belt, due to an increase in tension in the tense side of said Transmission Belt when an increased torque is being transmitted by said CVT. e) a Constrainer that removes and/or provides Transmission Belt Slack as needed when the axial position of said Driven Cone and said Driving Cone are aligned and not aligned for their current Transmission Ratio without substantially changing the angles at which said Transmission Belt approaches and leaves said Tensioning Pulley.
